International Rocket Engineering Competition
Design, Test, & Evaluation Guide

Revision: 2026 V1.1

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1.0 INTRODUCTION

The Experimental Sounding Rocket Association (ESRA) hosts and supports the International Rocketry Engineering Competition (IREC), a week-long series of events which will set the background and provide structure for the world's largest university rocket engineering competition. This new host event continues the IREC legacy of inspiring student design teams from across the country and around the world.

1.1 BACKGROUND

The smoke and fire, noise, high speeds, and sleek aerodynamics of rocketry encourage students to pursue science, technology, and mathematics-based careers. They have Rocket Fever! and competition motivates them to extend themselves beyond the classroom to design and build the rockets themselves. These students also learn to work as a team, solving real world problems under the same pressures they'll experience in their future careers.

ESRA held the first annual IREC in 2006. The competition achieved international status in 2011 when Canadian and Brazilian universities arrived. These schools have since been joined by others from every continent except Antarctica. The competition has increased in size every year since 2013, becoming the largest known collegiate level rocket engineering competition in the world. Attendance in 2016 included as many as 600 participants – including faculty, family, and friends of students from over 50 colleges and universities. The next year marked the start of a new era with the inaugural Spaceport America Cup. Over 1,100 students and representatives from 22 industry partners participated in an academic conference, rocket and payload engineering competitions, and non-competing demonstration flight tests. IREC continues to grow with nearly 200 teams applying for admission to the competition in 2023 with 121 teams and over 2,000 students from around the world making the trip to Spaceport America in June of 2023 for the competition.

In 2025, the competition transitioned to Midland, Texas with the strategic objective of inspiring students to elevate their engineering capabilities. This included the introduction of a 45,000-foot two-stage rocket category and a 100,000-foot waiver to push the boundaries of altitude achievement. The 2025 IREC marked a record-breaking year, drawing over 1,200 students to the event and engaging more than 7,500 participants across team rocketry programs, culminating in over 130 successful launches.

1.2 PURPOSE AND SCOPE

1.2.1 This document defines the minimum design, test, and evaluation criteria the event organizers expect teams to meet before launching at the IREC. The event organizers use these criteria to promote flight safety. Departures from the guidance this document provides may negatively impact an offending team's score and flight status, depending on the degree of severity. The foundational, qualifying criteria for the IREC are contained in the IREC Rules & Requirement document.

1.2.2 This document builds upon:

1.2.2.1 Tripoli Rocketry Association (TRA) Unified Safety Code (USC)

1.2.2.2 National Fire Protection Association (NFPA) Code for High Power Rocketry (NFPA 1127). While NFPA 1127, Section 1.3.3, exempts colleges and universities from its contents, these documents remain excellent supplemental resources for student researchers to learn more about best practices adopted by the amateur high-power rocketry community.

1.2.2.3 ESRA's observations on student launch initiatives.

1.2.2.4 Feedback received from student teams.

1.2.3 If any IREC team is unclear about competition rules and/or requirements, or has a situation not specifically addressed by the rules, they should contact ESRA by posting a message in the "dteg-questions" channel on Discord. Teams are encouraged to ask about both requirement interpretation and the spirit and intent of the rules and DTEG.

1.3 CONVENTION AND NOTATION

The following definitions differentiate between requirements and other statements. The degree to which a team satisfies the spirit and intent of these statements will guide the competition officials' decisions on a project's flight status and overall score at the IREC.

Shall: This is the only verb used to denote mandatory requirements. Failure to satisfy the spirit and intent of a mandatory requirement will always affect a project's score and flight status.

May: This verb signifies that teams are allowed, but not required, to exercise the associated action.

Should: This verb is used for stating non-mandatory goals. Failure to satisfy the spirit and intent of a non-mandatory goal may affect a project's score and flight status, depending on design implementation and the team's ability to provide thorough documentary evidence of their due diligence on-demand.

Will: This verb is used for stating facts and declarations of purpose. The authors use these statements to clarify the spirit and intent of requirements and goals.

1.4 Flight Status

Flight status refers to the granting of permission to attempt flight, and the provisions under which that permission remains valid. A project's flight status may be either nominal, provisional, or denied.

Nominal: A project assigned nominal flight status meets or exceeds the minimum expectations of this document and reveals no obvious flight safety concerns during flight safety review at the IREC.

Provisional: A project assigned provisional flight status generally meets the minimum expectations of this document but reveals flight safety concerns during flight safety review at the IREC which may be mitigated by field modification or by adjusting launch environment constraints. Launch may occur only when the prescribed provisions are met.

Denied: A project assigned Denied flight status has failed to meet the minimum expectations of this document or the Tripoli Unified Safety Code, or reveals unmitigated flight safety concerns. Competition officials reserve the right to deny flight status to any project deemed Denied.

1.5 Launch Vehicle versus Payload

An effort is made throughout this document to differentiate between launch vehicle and payload associated systems. Unless otherwise stated, requirements referring only to the launch vehicle do not apply to payloads and vice versa.

1.6 Revisions

The IREC DTEG may require revision periodically (from time to time or from one competition to the next), based on the lessons learned by both host organizations and the participants.

1.6.1 Major revisions will be accomplished by complete document reissue.

1.6.2 "Real world events" may require smaller revisions to this document in the months leading up to a competition.

1.6.3 Revisions will be reflected in updates to the document's effective date.

1.6.4 The authority to approve and issue revised versions of this document rests with ESRA.

2.0 Reference Documentation

The following documents include standards, guidelines, schedules, or required standard forms. The documents listed in this section are either applicable to the extent specified in this document or contain reference information useful in the application of this document.

DOCUMENT	FILE LOCATION
Location of Current IREC documents	https://www.esrarocket.org/documents-and-forms
IREC Rules & Requirements Document	https://www.esrarocket.org/s/IREC-Rules-and-Requirements-Document-2026-v10.pdf
IREC Integrated Master Schedule Document	https://www.esrarocket.org/s/2026-IREC-IMS-2026-V11-1.pdf
IREC Frequency Student Band Plan	https://www.esrarocket.org/s/2026-IREC-Student-Band-Plan-2026-V10.pdf
IREC Range Standard Operating Procedures[1]	https://www.esrarocket.org/s/irec_range_standard_operating_procedures_2025-a.pdf
IREC Live Rocket Video Challenge	https://www.esrarocket.org/live-rocket-video-challenge
IREC Entry Form & Progress Report	Obsoleted For The 2026 IREC As Entry Period Has Closed
TRA Unified Safety Code	https://www.tripoli.org/content.aspx?page_id=22&club_id=795696&module_id=520420
NFPA 1127: Code for High-Power Rocketry	https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=1127
14 CFR, Part 1, 1.1 General Definitions	http://www.ecfr.gov/cgi-bin/text-idx?SID=795aaa37494b6c99641135267af8161e&mc=true&node=se14.1.1_11&rgn=div8
14 CFR, Part 101, Subpart C, 101.22 Definitions	http://www.ecfr.gov/cgi-bin/text-idx?SID=795aaa37494b6c99641135267af8161e&mc=true&node=se14.2.101_122&rgn=div8
47 CFR, Part 97, Amateur Radio Service	https://www.ecfr.gov/current/title-47/chapter-I/subchapter-D/part-97

[1] The 2026 version of the IREC Range Standard Operating Procedures is being updated and will be made available before the IREC.

3.0 High Power Certified Flyer of Record

3.1 Tripoli Flight Requirements for All Competition Categories[2]

3.1.1 Every team shall engage a TRA Adult Flier, having a certification level appropriate for the motor being flown at IREC, to serve as their Flyer of Record (FoR). A Level 2 TRA Certification is required for K or L impulse motors or a combination of motors having a combined total impulse of less than 5,120 Ns. A Level 3 TRA Certification is required for M, N or O impulse motors or a combination of motors having a total impulse greater than 5,120 Ns. Any one of the following options shall satisfy this FoR requirement.

3.1.1.1 Recruit an L2 or L3 TRA-certified Adult Flier with significant rocketry experience to work closely with the team and then attend IREC as FoR. Student teams are encouraged to compensate or cover the travel expenses of their FoR.

3.1.1.2 If a team member, a student meets the requirement of 3.1.1.1. In this case, the student L2 team member can be the FoR.

3.1.1.3 Teams who have no access to local or regional Tripoli prefectures should first attempt a search using social media, forums, and other web-based sites (Facebook, Discord, *etc.*) to find a suitable FoR. If that search is unsuccessful, teams should contact the Tripoli Rocketry Association's Outreach Committee for assistance. (https://www.tripoli.org/outreach)

3.1.2 The FoR shall be present for the Fight Readiness Review (Horseshoe Pavilion), launch preparations, the Final RSO Check (immediately prior to entering queue), and for pad activities.

3.1.3 It is recommended that motors or energetic materials only be possessed or handled by TRA members with appropriate high power rocketry certification.

3.2 Team Members

3.2.1 Teams flying in the Solid or Hybrid categories, or the Demonstration category if using solid or hybrid motors, shall meet all TRA launch safety requirements as codified in the then-current Tripoli Rocketry Association Unified Safety Code.

3.2.2 All team members who will be onsite at the launch site shall hold a current TRA membership.

3.2.2.1 TRA Student membership shall satisfy requirement 3.2.2.

3.2.3 A maximum of 5 student team members may be on the pad loading team, although pad loading teams in the two-stage and hybrids area may be larger at the discretion of the pad manager.

3.2.3.1 The team Mentor may be present at the pads and does not count in the team limit in 3.2.3.

3.2.3.2 The FoR shall be present at the pads and does not count in the team limit in 3.2.3.

3.2.4 The recovery team shall contain a minimum of 2 and a maximum of 4 team members per 100 lbs. of recovery equipment, including the dry weight of the rocket. The Mentor

[2] A Tripoli L3-certified FoR is required for any category, including the demonstration category, that is allowed under Tripoli Rules. This excludes liquids.

and/or FoR may accompany the recovery team and are not included in the team member count.

4.0 Frequency Management And GPS Tracking Requirements

4.1 Student Radio Transmissions

4.1.1 All student radio transmissions shall meet the requirements of this section, including the IREC 2026 Student Band Plan (SBP).

4.1.2 The Student Band Plan (SBP) addresses the transmission characteristics for COTS GPS systems, SRAD transmissions and Video transmissions, and may address allowable transmission bands, frequency selection, forward RF power, transmission bandwidth (FWHM), spurious transmissions and harmonics.

4.1.2.1 The SBP may be updated from time to time or from one competition to the next, based on lessons learned by both ESRA and the participants. Revisions will be reflected in updates to the document's effective date. The authority to approve and issue revised versions of the SBP rests with ESRA.

4.1.2.2 Revisions of the SBP are not considered as revisions of this DTEG document.

4.1.2.3 The SBP is not an authorizing document. Participants are still fully responsible for complying with all applicable FCC regulations, regardless of anything stated in the plan.

4.1.2.4 The need for a US Amateur Radio license or operating authority are listed in the SBP.

4.2 COTS GPS Tracking Solutions

4.2.1 All independently recovered bodies shall include a COTS GPS tracking solution. Recommended COTS GPS trackers include the following:

Vendor	Product		Website
ESRA recommends the use of the 440 MHz (70 cm) band wherever possible			
Altus Metrum	TeleGPS, TeleMega, *etc.*	70 cm	https://altusmetrum.org/
Big Red Bee	Beeline GPS	70 cm	http://www.bigredbee.com/
Featherweight	Featherweight GPS	33 cm	https://www.featherweightaltimeters.com/featherweight-gps-tracker.html

4.2.2 COTS GPS systems from other suppliers may be used if they meet the requirements of this section, including the requirements listed in the IREC SBP.[3] The use of APRS protocol, as described in Section 1.1.1, is also recommended.

4.2.3 COTS GPS tracking solutions shall be switched and powered independently of any flight computer.

[3] At the present time, ESRA believes that the Silicdyne Fluctus and Entacore AIM Xtra units meet the requirements in the SBP. The Multitronics "Kate" unit can be used for IREC under the circumstances listed in the SBP.

4.2.3.1 COTS flight computers with integrated GPS tracking satisfy the requirements of both COTS GPS Tracking and Recovery Electronics and are exempt from (4.2.3) independent switch and power requirements.

4.3 Frequency Management

4.3.1 ESRA will be responsible for frequency management during the IREC.

4.3.1.1 ESRA will designate a specific range of frequencies (*i.e.*, COTS frequencies) within the 900 MHz (33 cm) band and within the 440 MHz (70 cm) band for the exclusive use of COTS GPS trackers.

4.3.1.2 COTS GPS trackers used within the COTS frequency ranges must meet the requirements of Section 4.1.2.

4.3.1.3 All RF sources must be documented in the required periodic Progress Reports or otherwise specified by ESRA, and must meet the requirements of Section 4.1.2.

4.3.2 For 2026, teams will be assigned specific frequencies for COTS GPS transmitters as follows:

4.3.2.1 Foreign teams from countries without reciprocal HAM license agreements will be assigned one or more COTS tracker frequencies within the 900 MHz (33 cm) band. Frequencies will be assigned by ESRA prior to the competition. Teams will surrender the use of these frequencies after completing the Post Flight Data Recovery process.

4.3.2.2 US teams, and teams from countries with reciprocal HAM license agreements, can elect to use COTS GPS trackers within the COTS GPS tracker range of the 440 MHz (70 cm) band. Teams selecting this frequency band will be assigned one or more COTS tracker frequencies by ESRA prior to the competition.

4.3.2.3 US teams, and teams from countries with reciprocal HAM license agreements, can elect to use COTS GPS trackers within the COTS GPS range of the 900 MHz (33 cm) band. Teams selecting this frequency band will be assigned one or more COTS tracker frequencies at the time of the **Final Flight Safety Review**. Note that teams will not be able to conduct their Final Flight Safety Review, nor test their GPS equipment, until the frequency or frequencies they need become available. Teams will surrender the use of these frequencies after completing the Post Flight Data Recovery process.

4.3.2.4 Teams using SRAD GPS systems or other SRAD transmitters shall use frequencies outside of the COTS tracker ranges in the 440 MHz (70 cm) and 900 MHz (33 cm) bands. These transmitters shall meet the requirements of Section 4.1.2.

4.3.2.5 Teams will not receive a frequency assignment for SRAD transmitters including SRAD GPS transmitters.

4.3.3 Teams with multiple independently recovered bodies can request separate frequencies for the COTS GPS device in each body or use time slotting (the preferred option). Multiple COTS GPS devices within a given body shall use time slotting.

4.3.4 With the exception of multiple independently recovered bodies, or time slotting of multiple COTS GPS transmitters, teams cannot use more than one frequency within the COTS GPS frequency ranges. For example, if a team has COTS GPS transmitters in both the 440 MHz and 900 MHz bands, one of those COTS GPS units will be assigned a frequency within the COTS GPS ranges and the other must operate within a designated SRAD frequency range.

4.3.5 Teams shall not power any transmitting electronics while in the pit area unless authorized by ESRA. This applies to transmitters both within and outside of the COTS frequency ranges within each band (i.e. all COTS and SRAD transmitters). Procedures for obtaining permission to test electronics will be finalized prior to the competition.

4.3.5.1 At the present time, it is anticipated that teams will be allowed to test their official COTS GPS units in the pit area after receiving their assigned operating frequencies (see 4.3.2 for specific information on the frequency assignment process for COTS GPS units).

4.3.5.2 No other radio transmissions will be allowed until the time of the Final Flight Safety review and as approved by the Queue Manager. Transmitters shall be powered off when not in use

4.3.5.3 Low power transmitters, as defined in the SBP, are allowed in the pit area. Transmitters shall be powered off when not in use.

4.3.6 Teams shall have the ability to quickly change frequencies on their COTS and SRAD transmitting and receiving equipment. If a team is directed to change frequencies, they will not be permitted to fly until such changes are completed and verified.

4.3.7 Teams shall not transmit on a COTS GPS frequency that is not assigned to them, or on any frequency that is not registered and approved by ESRA. Competition officials may assess penalties for teams that violate this requirement.

4.3.8 Trackers that allow custom names to be assigned to the tracker shall include the Team ID number. If the rocket contains multiple sections (e.g., the rocket and a payload or a two-stage booster and sustainer), the sections should be labeled. For example, Team 999's Section 1 tracker name would be "<T999S1>" (without the quotes). This is to aid in automatic parsing.

4.3.9 Should an SRAD transmission conflict develop between teams waiting in the queue or at the pads, it shall be the responsibility of the affected teams to resolve the conflict: Options for resolution of a conflict include the following:

4.3.9.1 One team changing their tracking system to a different (unused) frequency, or

4.3.9.2 One team delaying their launch to a subsequent salvo.

4.3.10 Transmitter operation associated with ground support equipment for Hybrids and Liquids shall coordinate their operations with the Hybrid Pad Manager.

4.3.11 Transmitter operation associated with the Live Video Challenge shall be consistent with the SBP and the rules associated with the 2026 Live Video Challenge

4.4 SRAD GPS Tracking Systems And Other SRAD Transmissions

4.4.1 SRAD GPS solutions may be used, at the option of the team, subject to the following:

4.4.1.1 Transmissions must be outside of the ranges designated for COTS GPS tracking and must meet the requirements listed in the SBP.

4.4.1.2 Additional documentation detailing the design, fabrication, operation, and testing of SRAD GPS tracking systems shall be included in progress reports.

4.4.2 APRS protocol via 1200 baud AFSK is recommended for SRAD GPS Tracking on the 2m, 70cm and 33cm bands, see http://www.aprs.org/doc/APRS101.PDF.

4.4.3 It is recommended to flight-test SRAD GPS systems.

4.4.4 Neither COTS GPS trackers nor SRAD GPS trackers will be used for official Post Flight Data Recovery altitude determination. Currently, altitude determinations are performed using barometric altitude as outlined in the Rules and Requirements document.

4.4.5 All SRAD transmissions shall follow the requirements in the SBP

4.5 FCC Amateur Radio Licensing

4.5.1 Teams are encouraged to get an Amateur Radio Operator's license for their country.

4.5.2 70cm and 2m APRS GPS Tracking solutions require a minimum of the control operator to be licensed at the Technician level or higher.

4.5.3 The control operator's call sign shall be used by the transmitter.

4.5.4 The control operator's call sign and team number shall be registered with ESRA via the periodic progress reports.

4.5.5 Teams utilizing Amateur Radio frequencies and based outside of the US shall secure US operating authority or use a solution transmitting at 900 MHz if a reciprocal licensing agreement cannot be obtained or cannot be used. US operating authority is most commonly accomplished via reciprocal licensing agreements.

4.5.6 International teams shall bring documentation in English (translated if necessary) of their US operating authority if using frequencies outside of 900 MHz.

4.6 GPS Tracking Solution Safety Inspections

4.6.1 IREC personnel at the pads shall verify that teams are able to communicate with their receiving station to confirm that their COTS GPS signals are being acquired and are functioning properly.

4.6.2 Launch operations will not be delayed because of COTS GPS inoperability. Teams which cannot confirm a COTS GPS Tracking signal shall not be allowed to launch.

4.6.3 The inoperability of an SRAD GPS will not be a reason to delay launch operations.

4.7 Recovery Team Training and Review at Team Registration

4.7.1 Prior to arrival Midland on Monday of the week of the IREC, all members of the recovery team shall have completed video training on utilization of the recovery backpack systems.

4.7.2 Teams may need to verify completion of training during the check-in process in Midland.

4.7.3 Teams must have a recovery plan before deployment. Teams that do not have good GPS information, or if the location is not known, must develop range and direction estimates in their recovery plan to avoid excessive wandering.

5.0 Propulsion Systems

5.1 Motor Categories

5.1.1 COTS Motors

5.1.1.1 A Commercial Off the Shelf (COTS) motor is defined as a motor which has been certified by the Canadian Association of Rocketry, Tripoli, or NAR, and appears on the then current combined Certified Rocket Motors List.

5.1.1.2 A list of all approved motors can be found on the TRA website: NAR-TMT Combined Motor List - Tripoli Rocketry Association

5.1.2 SRAD Motors

5.1.2.1 Student Research and Developed Motors (SRAD) are defined as

5.1.2.1.1 Any motor built in part or whole by students.

5.1.2.1.2 Any COTS solid motors modified in such a way as to conceivably affect operation or performance of the motor.[4]

5.1.2.2 Commercial solid propellant grains shall not be used in SRAD motors.

5.1.2.3 SRAD motors shall be static fired, and the propellant characterized prior to the submission of the Technical Report.

5.1.2.4 Static fire test results and propellant characterization data shall be presented during the Video Flight Readiness Review.

5.1.2.5 No second party motors (*i.e.*, not built by the team) shall be permitted.

5.2 Propulsion Types

5.2.1 Solid Motor

5.2.1.1 Any motor utilizing exclusively a solid propellant.

5.2.1.2 Solid propellants shall consist of non-toxic ingredients per Section 5.5.

5.2.1.3 SRAD solid motors shall be static fired, well characterized and tested per Section 5.16.

5.2.1.4 All SRAD solid motors shall comply with the TRA Unified Safety code.

5.2.2 Hybrid Motor

5.2.2.1 Any motor utilizing a combination of solid and liquid or gaseous propellants.

5.2.2.2 Hybrid propellants must be non-toxic as defined in Section 5.5.

5.2.2.3 SRAD hybrid motors shall be static fired, well characterized and tested per Section 5.16.

5.2.2.4 SRAD hybrid motors shall comply with the TRA Unified Safety code.

5.2.2.5 Nitrous oxide is the only allowable oxidizer.

5.2.3 Liquid Engine

5.2.3.1 An engine utilizing stored fuel and stored oxidizer in the liquid state. Nitrous Oxide is the only allowable Oxidizer for IREC.

5.2.3.2 All liquid propellants must be non-toxic as defined in Section 5.5.

5.2.3.3 Liquid engines shall be static fired, well characterized, and tested as per Section 5.16.

5.2.3.4 Liquid engines shall comply with the TRA Unified Safety Code.

[4] For example, modification of the forward closure of a motor to support head end ignition would result in the motor being classified as SRAD, as would any modification related to the threads used on the closures.

5.2.3.5 Nitrous oxide is the only allowable oxidizer.

5.3 Rail Exit Velocity

5.3.1 Teams shall design their rockets to attain a rail departure velocity of at least 25 m/s (82 ft/s) based on the "effective" length of the rail. "Actual" rail lengths are currently 17 feet for single-stage rockets and 21 feet for two-stage rockets.

5.3.2 The "effective" rail length is defined as the distance (altitude) that the rocket has traveled at the instant that the launch vehicle first becomes free to move about the pitch, yaw, or roll axis (i.e. the instant when only one rail button is connected to the rail).[5]

5.3.3 For tower launchers, the required rail departure velocity must be reached by the time that the center of pressure of the rocket has cleared the tower.

5.3.4 Cluster rockets shall meet the minimum rail exit velocity requirement on any single motor in the cluster.

5.4 Igniters

5.4.1 Teams shall supply the appropriate means of ignition for their motor.

5.4.2 Hybrid/Liquid teams shall discuss their igniter approach with ESRA safety personnel as soon as practicable.

5.4.3 For solid motors:

5.4.3.1 Ground-started motors shall be fired with dual-headed or dual-bridged igniters.

5.4.3.1.1 The use of two high-current igniters should be avoided due to excessive current requirements.

5.4.3.2 Motors may be started using either a COTS igniter or an SRAD igniter.

5.4.3.2.1 Teams shall specify the igniter(s) to be used during the Video Flight Readiness Review.

5.4.3.2.2 Teams using an SRAD igniter shall specify the design and composition of the igniter during the Video Flight Readiness Review.

5.4.3.2.3 Teams shall present their igniters for inspection at the Final Flight Safety Review.

5.4.3.3 COTS igniters that are modified shall be considered SRAD igniters and subject to the requirements of this section.

5.5 Non-Toxic Propellants

5.5.1 Launch vehicles entered in the IREC shall use non-toxic propellants.

5.5.2 Ammonium Perchlorate Composite Propellant (APCP), Potassium Nitrate and sugar (aka rocket candy), Nitrous Oxide, Kerosene, Propane, alcohol, and similar substances, are considered non-toxic.

5.5.3 Toxic propellants are defined as those requiring breathing apparatus, unique storage and transport infrastructure, or extensive personal protective equipment (PPE). Hydrogen Peroxide is considered a toxic propellant.

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[5] As an example, if a single-stage rocket has two rail buttons located 1foot and 4 feet from the bottom of the rocket, the effective rail length is 17 - 4 = 13 feet. The rail exit velocity is then the velocity of the rocket, as determined from the simulation of flight when the "altitude" of the rocket has reached 13 feet.

5.6 Sugar Motor Propellants

5.6.1 Sugar propellants shall use potassium nitrate as oxidizer.

5.6.2 The following sugars or sugar alcohols are allowable as fuel.

5.6.2.1 Dextrose

5.6.2.2 Erythritol

5.6.2.3 Sorbitol

5.6.3 Other chemicals may be used if the fuel includes more than 50% by weight of the sugars or sugar alcohols listed in 5.6.2.

5.7 Allowable materials for Case Components for SRAD Motors

5.7.1 Metallic cases shall be made of non-ferrous ductile metals such as 6061 aluminum alloy.

5.7.2 Non-metallic cases shall not be made of brittle materials which may rupture into sharp shards, such as PVC or other low-temperature polymers.

5.7.3 Forward closures shall not be made of ferrous materials.

5.7.4 Minor components such as snap rings, nozzle washers, rear closures, and seal disks may be made of ferrous metals.

5.7.5 Nozzles for Sugar Motors may be made of steel as long as the throat of the nozzle is recessed within the case.

5.7.6 Combustion chambers for hybrid motors shall not be made from any alloy of steel.

5.8 Propulsion System Safing and Arming

5.8.1 A propulsion system shall be deemed armed if only one action (*e.g.*, an ignition signal) must occur for the propellant(s) to ignite.

5.8.2 The “arming action” is any action enabling an ignition signal to ignite the propellant(s).

5.8.3 The IREC provided launch control system described in Section 11.2 will be used for all solid and two-stage launches.

5.8.4 Hybrid or liquid teams shall provide launch control systems that ensure their rocket is properly made safe, *i.e.*, not armed until commanded. Additional requirements for team-provided launch control systems are defined in Section 11.3 of this document.

5.9 Propellant Offloading After Launch Abort

5.9.1 Hybrid and liquid propulsion systems shall implement a means for remotely controlled venting or offloading of all liquid and gaseous propellants.

5.9.2 Normally open purge valves shall be used.

5.10 Ground-Start Ignition Circuit Arming

5.10.1 All personnel shall be at least 15 m (50 ft) away from the launch vehicle prior to ground-started propulsion system ignition circuits/sequences being armed.

5.10.2 The IREC provided launch control system satisfies this requirement.

5.11 AIR-START/STAGED IGNITION CIRCUIT ARMING

5.11.1 All upper-stage (*i.e.*, air-starts) propulsion systems shall be designed to:

5.11.1.1 Prevent motor ignition during arming on the ground.

5.11.1.2 Inhibit motor ignition, as defined in Section 5.13, in the event of a non-nominal flight.

5.11.1.3 Provide the capability of being disarmed in the event the rocket is not launched.

5.12 AIR-START/STAGED IGNITION CIRCUIT ELECTRONICS

5.12.1 All upper-stage ignition systems shall comply with either the same requirements for Safety Critical Wiring as recovery systems as defined in Section 6.12, or be specifically approved by the Two-Stage Pad Manager.

5.13 AIR-START/STAGED FLIGHT COMPUTER REQUIREMENTS

5.13.1 Ignition of air-start motors shall be accomplished using a COTS flight computer configured to inhibit ignition of the air-start motor(s) unless all the following conditions have been met:

5.13.1.1 Booster burnout has been detected.

5.13.1.2 The rocket has reached an altitude of at least 70% of the simulated altitude at the time when initiator firing is desired (the simulation shall be performed at a 0° launch angle with no wind).

5.13.1.3 The sustainer flight angle is within 20° of vertical.

5.13.2 The flight computer shall be configured to prevent the air-start motor from firing at a later time if the altitude threshold is not achieved.

5.13.3 Currently available flight computers that satisfy the requirements of this section include, but are not limited to:

5.13.3.1 Featherweight Blue Raven and Blue Jay Plus

5.13.3.2 Aim Xtra

5.13.3.3 Multitronix Kate

5.13.3.4 Altus Metrum Telemega and EasyMega

5.13.3.5 MARSA Systems Marsa33 and Marsa54 flight computers with Tilt Module & Interface.

5.13.4 SRAD flight computers shall not be used for the purpose of igniting air-start motors.

5.13.5 Simple timers shall not be used for igniting air-start motors unless in combination with altitude and tilt lockout.

5.14 AIR-START/STAGED FLIGHTS – ARMING PROCEDURES

5.14.1 Projects shall include provisions that prevent air-start motor ignition on the ground.

5.14.2 All projects shall include a provision to open the circuit between the flight computer and the air-start motor igniter, and to shunt the igniter, prior to and during power-up of the flight computer.

5.14.2.1 The requirements in Section 5.14.2 can be accomplished using one or more than one physical switch.

5.14.2.2 Examples of recommended switch designs are provided in Appendix D.

5.14.2.3 The Multitronix Kate flight computer meets the shunt requirement of Section 5.14.2.

5.14.2.4 The use of SRAD electronics for the purpose of remotely accomplishing the switching functions (isolation and shunt) in this section can be considered; however, under no circumstances can such electronics cause a current to flow to the igniter.

5.14.3 Flight computers shall not be powered up until the rocket is in a vertical position.[6]

5.14.3.1 Devices that will power flight computers via remote operation, such as Wi-Fi switches, can be powered up prior to lifting the rocket to a vertical position.

5.14.4 The electronics configuration shall be designed such that the provisions used to open and to shunt the circuit to the igniter can be used to safe the igniter if the rocket is not launched.

5.15 AIR-START/STAGED FLIGHTS – ADDITIONAL INFORMATION REQUIREMENTS

5.15.1 Teams shall provide additional information in the second progress report specifically related to air-start/staged flights.

5.15.2 Updated information, including a response to ESRA comments as applicable, shall be included in the third progress report and discussed during the Video Flight Safety Review.

5.15.3 The required additional information includes:

5.15.3.1 Schematic diagram of the electronics configuration to be used for air-start motor ignition and recovery.

5.15.3.2 A graph illustrating flight simulation profile, to include altitude, velocity, and acceleration as a function of time to the expected apogee time.

5.15.3.3 An explanation of the strategy for the flight based on the above flight profile (i.e., what is the rationale behind the selection of staging times, coast times, etc.)

5.15.3.4 A description of the specific procedures that will be used to prevent air-start motor ignition on the ground.

5.15.3.5 A description of the specific procedures that will be used to inhibit air-start motor ignition in the event of a non-nominal flight.

5.15.3.6 Drawing & description of the interstage coupler.

5.15.3.7 Drawing and description of the wiring design between the flight computer and the igniter, including any break wires employed on this wiring.

5.16 SRAD PROPULSION SYSTEM TESTING

5.16.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

5.16.2 Teams shall complete required testing of SRAD and modified COTS propulsion systems prior to submitting their Technical Report. Teams unable to make this deadline may switch to COTS but must do so prior to submitting their Technical Report.

5.16.3 Combustion Chamber Pressure Testing

[6] Teams are encouraged to develop processes that allow the power-up or arming procedures in this section to be conducted from the ground rather than from a ladder.

5.16.3.1 SRAD and modified COTS propulsion system combustion chambers shall be designed and tested according to the SRAD pressure vessel requirements defined in Section 6.18 of this document.

5.16.3.2 Combustion chambers are exempted from the requirement for a relief device.

5.16.4 Hybrid and Liquid Propulsion System Tanking Testing

5.16.4.1 SRAD and modified COTS propulsion systems using liquid or hybrid propellant(s) shall successfully complete a propellant loading and off-loading test in “launch-configuration”.

5.16.4.2 This test may be conducted using either actual propellant(s) or suitable proxy fluids.

5.16.4.3 The loading and off-loading test shall demonstrate meeting the fill-to-fire time and detanking time requirements of 5.17.

5.16.4.4 Links to videos and testing data shall be included with, or linked to, in the third progress report and final report.

5.16.5 Static Hot-Fire Testing

5.16.5.1 SRAD propulsion systems shall successfully complete an instrumented (chamber pressure and thrust), full scale (including system working time) static hot-fire test prior to the IREC.

5.16.5.2 SRAD Solid teams shall provide a BurnSim or openMotor file of their intended motor.

5.16.5.3 SRAD Solid teams shall provide static test data including at least a graph of the pressure and thrust versus time.

5.16.5.4 SRAD static test results shall be provided in the Comma Separated Values (CSV) file format and include Time (sec), Pressure (pascals) and Thrust (newtons) data.

5.16.5.5 Video of successful testing shall be included with, or linked to, in the third progress report and final report.

5.16.5.6 The flight motor shall represent the static test motor in all respects.

5.16.5.7 Any changes to the SRAD flight motor shall be tested and resubmitted.

5.17 SRAD Hybrid/Liquid Pad Operations Design Requirements

5.17.1 Fill to fire time shall be ≤30 minutes. The filling time is defined as the time from filling start to launch ignition.

5.17.2 Pad managers may, using their discretion, permit an additional 15 minutes of fill time.

5.17.3 Rockets shall vent/detank in ≤30 minutes, including in fault and failsafe conditions.

5.17.4 Rockets shall be able to hold fully filled for ≥10 minutes before launching.

5.17.5 Teams will be told to abort if they do not meet these times.

6.0 Recovery Systems and Avionics

6.1 Dual-Event Parachute and Parafoil Recovery

6.1.1 Each independently recovered launch vehicle body, including payloads, anticipated to reach an apogee above 457 m (1,500 ft) above ground level (AGL) shall follow a "dual-event" recovery operations concept (CONOPS) defined to include the following:

6.1.1.1 A drogue deployment event of a drogue parachute (or a reefed main parachute) at or near apogee.

6.1.1.2 A main deployment event of a main parachute, or unreefing of a main parachute, at an altitude no higher than 457 m (1,500 ft).

6.1.2 Independently recovered bodies (payloads) released at an altitude below 1,500 ft (457 m) AGL, are exempted from dual deployment and may feature only a single/main deployment event, which must occur at or near the point of release.

6.1.3 Boosters from two-stage rockets are subject to the dual-event provisions of this section.

6.1.4 A single commercially available release device appropriately rated for large parachutes, *e.g.*, Tinder Rocketry Tender Descender L1, L2, L3, and TD-2, the Rattworks AARD, *etc.*, are permitted subject to the following:

6.1.4.1 Dual initiators shall be used.

6.1.4.2 Redundant energetics (powder charges) are not required.

6.1.4.3 Initiators shall be activated by independent flight computers.

6.1.4.4 The device shall be successfully ground tested under a load approximating the load anticipated during flight.

6.1.4.5 The Jolly Logic Chute Release (JLCR) is not permitted for any recovery purpose.

6.1.5 Commercially available line/cable cutters, *e.g.*, Tinder Rocketry Piranha and Mako, Archetype Cable Cutter, *etc.*, are permitted subject to the following:

6.1.5.1 Redundant devices, activated by independent flight computers, shall be used.

6.1.5.2 The use of dual initiators in a single device does not satisfy 6.1.5.1.

6.1.6 SRAD release devices and line/cable cutter devices are permitted subject to the following:

6.1.6.1 Redundant devices shall be used (for both release and cable cutter devices).

6.1.6.2 Activation of redundant devices shall be controlled by independent flight computers.

6.1.6.3 The SRAD device shall be successfully ground tested under a load approximating the load anticipated during flight.

6.1.7 No portion of the rocket or payload can be "guided" during any phase of the recovery process. The term "guided" is intended to include, but is not limited to, the use of GPS or other external sensor input, whether this input comes from on board the rocket or from the ground.

6.2 Recovery Area

6.2.1 Any hardware drifting outside the safe recovery area or onto properties with restricted access will either be abandoned or recovered at the team's own expense.

6.2.2 Range Safety personnel may deem a recovery design as unsafe if it is determined that there is a reasonable probability the recovery could depart the safe recovery area.

6.3 Initial Deployment Event

6.3.1 The initial deployment event shall occur at or near apogee.

6.3.2 Descent rate calculations shall be based on a launch site elevation of 890 m (2,920 ft) MSL.

6.3.3 The descent rate under drogue shall be between 20-40 m/s (65-131 ft/s) at the point where the main parachute is deployed.

6.4 Main Deployment Event

6.4.1 The main deployment event for any recovery method shall occur at an altitude no higher than 457 m (1,500 ft) AGL.

6.4.2 Descent rate calculations shall be based on a launch site elevation of 890 m (2,920 ft) MSL.

6.4.3 The vehicle's descent rate at landing shall be less than 11 m/s (36 ft/s).

6.5 Ejection Gas Protection

6.5.1 The recovery system shall implement adequate protection (*e.g.*, fire resistant material, pistons, baffles, *etc.*) to prevent ejection energetics from damaging recovery components including, but not limited to, shock cords and parachutes.

6.6 Parachute Swivel Links

6.6.1 The recovery system rigging (*e.g.*, parachute lines, risers, shock cords, *etc.*) may implement appropriately rated swivel links at connections to relieve twist/torsion as the specific design demands.

6.7 Parachute Coloration and Markings

6.7.1 When separate parachutes are used for the initial and main deployment events, these parachutes shall be contrasting colors: from each other, blue sky, and clouds.

6.7.2 Through the use of visually distinct parachute colors and/or patterns, teams should be able to visually determine if a main was deployed at apogee, or was not deployed at the expected deployment altitude during descent (drogue only recovery).

6.8 Non-Parachute/Parafoil Recovery Systems

6.8.1 Teams utilizing non parachute or parafoil-based recovery methods shall notify ESRA of their intentions at the earliest possible opportunity and keep ESRA apprised of their progress.

6.8.2 ESRA reserves the right to impose additional requests for information and draft unique requirements depending on the team's specific design.

6.9 Redundant Electronics

6.9.1 Launch Vehicles shall implement completely independent and redundant recovery systems to include arming switches, sensors/flight computers, power supplies, redundant energetics (except as noted in Section 6.1.4), and electric initiators.

6.9.2 A dual pole switch does not satisfy the requirement for redundant arming switches.

6.10 REDUNDANT COTS RECOVERY ELECTRONICS

6.10.1 At least one of the redundant recovery subsystems in each independently recovered section of the rocket shall implement a COTS flight computer (*e.g.*, StratoLogger, AIM, Kate, EasyMini, TeleMetrum, RRC3, Raven, *etc.*).

6.10.2 A COTS flight computer shall serve as the official altitude logging system as specified in Section 2.5 of the IREC Rules & Requirements Document.

6.10.2.1 This function may be provided by one of the flight computers satisfying 6.9.1 or a separate flight computer.

6.10.2.1.1 A separate flight computer may be armed and powered by the same switch and power supply as one of the flight computers satisfying 6.9.1.

6.10.3 A COTS flight computer shall fire either the primary or redundant energetic system.

6.10.4 To be considered COTS, the flight computer (including flight software) shall have been developed and validated by a commercial third party.

6.10.5 Commercially designed flight computer kits (*e.g.*, the Eggtimer or similar) are not permitted as COTS.

6.11 SRAD RECOVERY ELECTRONICS

6.11.1 Flight computer kits, such as Eggtimer or similar may be used as SRAD electronics.

6.11.2 Any SRAD flight computer assembled from separate COTS components shall be considered an SRAD system.

6.11.3 A COTS microcontroller running SRAD flight software shall be considered an SRAD system.

6.11.4 A COTS flight controller containing any student modifications to the base software or hardware shall be considered an SRAD system.

6.11.5 SRAD flight computers shall be documented in the progress reports.

6.11.6 Teams shall provide proof of function of SRAD flight computers, including ground testing and flight testing before the competition.

6.11.7 SRAD recovery electronics shall take appropriate recognition of effects of noise and sensor error. Use of appropriate sensor filtering with testing is required.

6.12 SAFETY CRITICAL WIRING FOR RECOVERY SYSTEMS AND AIR-START MOTORS

6.12.1 Definition

6.12.1.1 Safety critical wiring is any electrical wiring associated with recovery system deployment events, air start motors, location beacons (*e.g.*, GPS), and systems that can affect the rocket trajectory or stability.

6.12.1.2 All safety critical wiring shall meet the requirements in this section and the safety critical wiring guidelines described in Appendix B of this document.

6.12.2 Cable Management

6.12.2.1 All safety critical wiring shall implement a cable management solution (*e.g.*, wire ties, wiring, harnesses, cable raceways) which will prevent tangling and excessive free movement of significant wiring/cable lengths due to expected launch loads.

6.12.2.2 Wiring shall be labeled or color-coded to allow easy determination of the subsystem to which it is attached.

6.12.2.3 Wiring shall incorporate strain relief at all connections/terminals to prevent unintentional de-mating/disconnection.

6.12.3 Secure Wiring Connections

6.12.3.1 All safety critical wiring/cable/terminal connections shall be sufficiently secure as to prevent disconnection during handling, transport, launch preparation and flight.

6.12.3.2 Electrical connectors shall not be used as structural elements, *e.g.*, the mounting of daughterboards on a mainboard.

6.12.3.3 Structural supports (*e.g.*, screws and standoffs), are required for daughterboard mounting.

6.12.3.4 The use of twisted wires, even with the use of wire nuts or similar devices, to make or break flight essential connections is prohibited.

6.13 Recovery System Testing

6.13.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

6.13.2 The following requirements concern verification testing of all recovery systems.

6.13.3 All recovery system ground tests should be completed by the Video Flight Readiness Review and shall be completed prior to flight.

6.13.4 Ground test demonstration

6.13.4.1 All recovery system mechanisms shall be successfully tested, without significant anomalies, prior to the IREC through one or more ground tests of key subsystems.

6.13.4.2 Sensor electronics shall be functionally included in ground test demonstrations by simulating the environmental conditions under which their deployment functions are triggered (*i.e.*, expected ambient atmospheric conditions).[7]

6.13.4.3 Video(s) of ground test demonstrations shall be included, or linked to, in the third progress report. Note, however, that "pop" testing is allowed at the competition site and proof of successful pop testing, without anomalies, is required prior to flight.

6.13.5 Flight Test Demonstration

6.13.5.1 A flight test demonstration is recommended and may be required by flight safety personnel in the case of an unusual or untested recovery approach.

6.13.5.2 In the case of such a flight test, the recovery system flown shall verify the intended design by implementing the same major subsystem components (*e.g.*, flight computers and parachutes) as will be integrated into the launch vehicle intended for the IREC.

6.13.5.3 Video(s) of the test flight(s) shall be included, or linked to, in the third progress report.

[7] This requirement means that sensor electronics should be ground tested at the conditions they will experience during flight. It does not require that a flight computer itself must be used to trigger a recovery charge during a ground test (i.e., a pop test).

6.14 Stored-Energy Devices – Energetic Device Safing and Arming

6.14.1 For the purposes of this section energetics are defined as all stored-energy devices, other than propulsion systems, that have reasonable potential to cause bodily injury upon energy release.

6.14.2 An energetic device is considered made safe when two separate events are necessary to release the energy.

6.14.3 An energetic device is considered armed when only one event is necessary to release the energy.

6.14.4 All energetics shall be made safe until the rocket is in the launch position, at which point they may be armed at the direction of IREC personnel.

6.14.5 The following table lists some common types of stored-energy devices and overviews in what configuration they are considered non-energetic, made safe, or armed.

DEVICE CLASS	NON-ENERGETIC	MADE SAFE	ARMED
Igniters/Squibs	Small igniters/squibs, nichrome, wire or similar	Large igniters with leads shunted	Large igniters without shunted leads
Pyrogens (*e.g.,* black powder)	Very small quantities contained in non-shrapnel producing devices (*e.g.,* pyro-cutters or pyro-valves)	Large quantities with no igniter, shunted igniter leads, or igniter(s) connected to unpowered avionics	Large quantities with non-shunted igniter or igniter(s) connected to powered avionics
Mechanical Devices (*e.g.,* powerful springs)	De-energized/relaxed state, small devices, or captured devices (*i.e.,* no jettisoned parts)	Mechanically locked and not releasable by a single event	Unlocked and releasable by a single event
Pressure Vessels	Non-charged pressure vessels	Charged vessels with two events required to open main valve	Charged vessels with one event required to open main valve

6.14.6 Teams shall not bring amounts of pyrogenic materials to the event greater than the amount required for ground testing and the competition flight.

6.14.7 All stored-energy devices (aka energetics) used in recovery systems shall comply with the energetic device requirements defined in this section.

6.15 Arming Devices

6.15.1 All energetic device arming features shall be externally actuatable and controllable. Magnetic switches and the Multitronix turn-on coil shall be understood to be externally actuatable and controllable.

6.15.2 Access panels which may be secured for flight while the vehicle is in the launch position are acceptable.

6.15.3 All energetic device arming features shall be accessible without any person placing their head against the rocket or within 100 mm of the rocket.

6.15.4 All energetic device arming features shall be operable while wearing a face shield and any other required PPE.

6.15.5 All energetic device arming features shall be located on the airframe such that any inadvertent energy release will not impact personnel arming them.[8]

6.15.6 Teams are responsible for verifying that they will be able to reach their arming switches. Currently, rockets in the solids area are supported approximately 2 feet above the ground with (at least) 10-foot stepladders available. In the two-stage area, rockets are supported approximately 4 feet above the ground with 20-foot extension ladders available.

6.16 Arming Device Verification

6.16.1 Arming shall be verified without any person placing their head against the rocket or within 100 mm of the rocket. A stethoscope may be used to increase the distance between an audio source and a human ear.

6.16.2 Arming verification shall be designed to be completed in an environment with at least 76 dBA background noise plus at least 32 km/h (20 mph) wind.

6.16.3 Arming verification shall be designed to be completed while wearing a face shield and any other required PPE.

6.17 Batteries Contained In The Rocket

6.17.1 Prohibited Batteries

6.17.1.1 Lithium-Polymer (LiPo) batteries are not permitted due to fire hazard.

6.17.1.2 Li-Ion batteries in a rectangular form factor and/or with plastic casing are prohibited.

6.17.1.3 Wet cell lead acid batteries are prohibited.

6.17.1.4 Any nuclear batteries

6.17.2 Allowed Batteries

6.17.2.1 LiFePO4 chemistry cells are allowed in any form factor and in any casing (plastic or metallic).

6.17.2.2 NiMH (Nickel-Metal hydride) batteries are allowed in metal casing and any form factor.

6.17.2.3 Alkaline (non-rechargeable) batteries are allowed in metal casing and any form factor.

6.17.2.4 Other Li-Ion batteries are permitted if packaged in a cylindrical metallic casing.

6.17.2.5 Lithium primary coin cell batteries in metal casing are permitted.

6.17.3 Teams should inquire about battery chemistries and form factors not listed here.

[8] For example, the arming key switch for an energetic device used to deploy a hatch panel shall not be located at the same airframe clocking position as the hatch panel deployed by that charge.

6.18 SRAD Pressure Vessels

6.18.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

6.18.2 The following requirements concern design and verification testing of SRAD and modified COTS pressure vessels.

6.18.2.1 Unmodified COTS pressure vessels utilized for other than their advertised specifications will be considered modified, and subject to these requirements.

6.18.3 Relief Device

6.18.3.1 All SRAD pressure vessels shall implement a relief device, set to open at no greater than the proof pressure specified in the following requirements.

6.18.3.2 SRAD (including modified COTS) rocket motor propulsion system combustion chambers are exempted from this requirement.

6.18.4 Designed burst Pressure for Metallic Pressure Vessels

6.18.4.1 All SRAD and modified COTS pressure vessels constructed entirely from isotropic materials (*e.g.*, metals) shall be designed to a burst pressure no less than 2 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.

6.18.5 Designed Burst Pressure for Composite Pressure Vessels

6.18.5.1 All SRAD and modified COTS pressure vessels either constructed entirely from non-isotropic materials (*e.g.*, fiber reinforced plastics; FRP; aka composites), or implementing composite overwrap of a metallic vessel (aka composite overwrapped pressure vessels; COPV), shall be designed to a burst pressure no less than 3 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.

6.18.6 SRAD Pressure Vessel Testing

6.18.6.1 Proof Pressure Testing – SRAD and modified COTS pressure vessels shall be proof pressure tested successfully to 1.5 times the maximum expected operating pressure for no less than twice the maximum expected system working time. Pressurized water (hydrostatic) testing is recommended unless other ‘as safe' methods are available. The intended flight article(s), that is, the pressure vessel(s), used in proof testing must be the same one(s) flown at the IREC).

6.18.6.2 Burst Pressure Testing

6.18.6.2.1 There is no requirement for burst pressure testing, however a rigorous verification & validation test plan typically includes a series of both non-destructive (*i.e.,* proof pressure) and destructive (*e.g.*, burst pressure) tests.

6.18.6.2.2 A series of burst pressure tests performed on the intended design will be viewed favorably; however, this will not be considered an alternative to proof pressure testing of the intended flight article.

7.0 Active Flight Control Systems

7.1 Restricted Control Functionality

7.1.1 All launch vehicle active flight control systems shall be implemented strictly for pitch and/or roll stability augmentation, or for aerodynamic braking.

7.1.2 Under no circumstances will a launch vehicle entered in the IREC be actively guided towards a designated spatial target or GPS location.

7.1.3 ESRA reserves the right to make additional requests for information and draft unique requirements depending on the team's specific design.

7.2 Unnecessary For Stable Flight

7.2.1 Launch vehicles implementing active flight controls shall be naturally stable without those controls being implemented (*e.g.*, the launch vehicle may be flown with the control actuator system (CAS) – including any control surfaces – either removed or rendered inert and mechanically neutral, without becoming unstable during ascent).

7.2.2 Attitude control systems (ACS) shall only mitigate small perturbations which affect the trajectory of a stable rocket that implements only fixed aerodynamic surfaces for stability.

7.2.3 Stability is defined in Section 10.2 of this document.

7.2.4 ESRA reserves the right to make additional requests for information and draft unique requirements depending on the team's specific design.

7.3 Designed To Fail Safe

7.3.1 Control actuator systems (CAS) shall default to a neutral state whenever either an abort signal is received for any reason, primary system power is lost, or the launch vehicle's attitude exceeds 30° from its launch elevation.

7.3.2 A neutral state is defined as one which does not apply any moments to the launch vehicle (*e.g.*, aerodynamic surfaces trimmed or retracted, gas jets off, *etc.*).

7.4 Boost Phase Dormancy

7.4.1 Control actuator systems (CAS) shall remain in a neutral state until one of the following conditions is met:

7.4.1.1 The launch vehicle's boost phase has ended (*i.e.*, all propulsive stages have ceased producing thrust).

7.4.1.2 The launch vehicle has crossed the point of maximum aerodynamic pressure (max Q) in its trajectory.

7.4.1.3 The launch vehicle has reached the following:

7.4.1.3.1 For 30K and 45K flights: an altitude of 6,000 m (19,600 ft) AGL.

7.4.1.3.2 For 10K flights: an altitude of 2,000 m (6,500 ft) AGL.

7.5 Active Flight Control System Electronics

7.5.1 All active control systems shall comply with the "safety critical wiring" requirements, 6.12.

7.5.2 Flight control systems are exempt from the requirement for COTS redundancy, given that such components are generally unavailable as COTS to the amateur high-power rocketry community.

7.6 ACTIVE FLIGHT CONTROL SYSTEM ENERGETICS

7.6.1 All stored-energy devices used in an active flight control system (aka energetics) shall comply with the energetic device requirements defined in Section 6.14 of this document.

8.0 AIRFRAME STRUCTURES

8.1 ADEQUATE VENTING

8.1.1 Launch vehicles shall be adequately vented to prevent internal pressures developing during flight and causing either damage to the airframe or any other unplanned configuration changes.

8.1.2 Teams shall discuss airframe and altimeter bay venting as part of their Video Flight Readiness Review.

8.2 OVERALL STRUCTURAL INTEGRITY

8.2.1 Launch vehicles shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling and transportation and during rocket flight.

8.2.2 Teams shall ensure that the fin flutter velocity of the rocket is at least 50% higher than the maximum expected rocket velocity.

8.2.3 Teams shall provide a value for the shear modulus of the fin material if that value is used in the calculation of the fin flutter velocity. Values may be estimated, based on supplier information, or determined from actual measurements.

8.3 MATERIAL PROHIBITIONS

8.3.1 Rockets shall be built using lightweight materials, *e.g.*, fiberglass and carbon fiber, or when necessary ductile lightweight metals, *e.g.*, aluminum, and construction techniques suitable for the planned flight. Teams should refer to the Tripoli Unified Safety Code, and specifically, to "additional information" presented on the Tripoli Website regarding the use of metal in rocket construction.

8.3.2 PVC (and similar low-temperature polymers), Public Missiles Ltd. (PML) Quantum Tube, and steel (including stainless-steel) shall not be used in any rocket airframe structure. This includes fins, body tubes, and nosecones.

8.3.3 PVC (and similar low-temperature polymers), stainless steel, or other frangible materials shall not be used for a solid or hybrid propulsion system combustion chamber.

8.3.4 Environmentally hazardous materials shall not be used as ballast. This includes lead, mercury, and uranium.

8.4 LOAD BEARING EYE BOLTS AND U-BOLTS

8.4.1 All load bearing eye bolts shall be of the closed-eye, forged type.

8.4.2 All load bearing eye bolts, U-bolts, and links shall be steel.

8.4.3 This requirement extends to any bolt and eye-nut assembly used in place of an eyebolt.

8.4.4 Stainless steel components (eye bolts, U-bolts, links, *etc.*) are permissible for use in recovery systems.

8.5 JOINTS IMPLEMENTING COUPLING TUBES

8.5.1 Airframe-to-coupler sliding joints intended to separate during a recovery event

8.5.1.1 Joints shall be designed such that the coupling tube extends no less than 1 body tube diameter (1 caliber) into the airframe section from which the coupler will separate during flight.[9]

8.5.2 Joints not intended to separate during flight

8.5.2.1 Joints shall be designed such that the coupling tube extends into the mating component to the lesser of 1 body tube diameter (1 caliber) or the maximum depth possible by the design of the mating components.

8.5.2.2 Joints shall be affixed by mechanical fasteners and/or permanent adhesive.

8.5.3 Regardless of implementation (*e.g.*, RADAX or other joint types) airframe joints shall prevent bending, see https://www.osti.gov/biblio/5007820.

8.6 Rail Buttons

8.6.1 Rail buttons shall implement “hard points” for sliding mechanical attachment of the rocket to the IREC supplied 1515 launch rail, serving to guide the rocket during the initial phase of boost until the rocket achieves sufficient velocity for the fins to provide aerodynamic stabilization.

8.6.2 A minimum of two (2) rail buttons shall be used, and the locations of the rail buttons shall be clearly indicated in the OpenRocket simulation required in progress reports.

8.6.3 Airframe attachment points for rail buttons shall be reinforced.

8.6.4 Rail buttons shall be attached using at least one metallic fastener through the reinforced airframe.

8.6.5 Adhesive-only attachment is not permitted.

8.6.6 Fly-away rail guides are not permitted.

8.6.7 Rail buttons 3D printed in a polymer material are not permitted.

8.6.8 Conformal, adhesively attached rail guides of a linear shape sliding in the slot in the launch rail are not permitted.

8.6.9 The rail buttons shall be capable of supporting the launch vehicle’s fully loaded weight while the rocket is horizontal and hung from a rail (underslung). The attachment procedure and strength of the attachment will be evaluated during the Horseshoe Pavillion Flight Safety Review.

8.6.10 Rail button placement shall not result in the rail blocking access to arming electronics and should also not interfere with disassembling the rocket.

8.6.11 Compliance with the above requirements may not be possible for minimum diameter rockets. In such cases, the rail button attachment approach shall be documented in progress reports and is subject to ESRA approval.

8.6.12 Teams utilizing launch towers shall provide the launch tower and are exempt from the Rail Button requirements.

[9] For the 2026 IREC, a sliding joint of less than one caliber will be allowed between a COTS nose cone and the nose cone coupler. This exception is to allow the so-called Head-End Dual Deploy (HEDD) approach. The overlap must exceed at least one-half of the body tube diameter (1/2 caliber) and the coupler tube must be inserted to the maximum depth allowed by the design of the nose cone.

8.7 Shear Pins

8.7.1 Teams shall indicate the locations of shear pins on drawings or simulations of their rocket.

8.7.2 Teams shall describe the number and size of shear pins used in various sections of the rocket, and how this was determined, as part of the recovery system discussion in Progress Report 2.

8.8 Identifying Markings

8.8.1 All rockets shall be labeled with the school name, the Team ID number, and the assigned GPS tracking frequency.

8.8.1.1 This label shall be duplicated on each part of the rocket which could separate either as designed or accidentally.

8.8.1.2 There are no specific size or format requirements for these labels.

8.8.2 The booster airframe shall prominently display the Team ID number forward of each fin. The digits of the Team ID shall be marked in sequence, longitudinally, from fore to aft along the booster airframe. See Appendix F for an example.

8.8.2.1 Individual digits shall be oriented with their vertical axis in-line with the longitudinal axis of the rocket and shall be readable with the rocket in a vertical position.

8.8.2.2 The Team ID digits shall each be at least 75 mm (3 in) high, at least 12 mm (½ in) stroke width, and a solid color that clearly contrasts with a solid background color.

8.8.2.3 The Team ID number should be clear to livestream viewers when the rocket is on the pad.

8.9 Other Markings

8.9.1 There are no requirements for airframe coloration or markings beyond those specified in Section 8.8 of this document; however, ESRA offers the following recommendations to student teams.

8.9.1.1 Mostly white or lighter tinted color (*e.g.*, yellow, red, orange, *etc.*) airframes are especially conducive to mitigating some of the solar heating experienced in the IREC launch environment.

8.9.1.2 High-visibility schemes (*e.g.*, high-contrast black, orange, red, *etc.*) and roll patterns (*e.g.*, contrasting stripes, *etc.*) may allow ground-based observers to more easily track and record the vehicle's trajectory with high-power optics.

8.9.1.3 Any form of green or brown, or other colors associated with camouflage patterns, are highly discouraged.

9.0 PAYLOAD

9.1 PAYLOAD RECOVERY

9.1.1 Payloads may be deployable independent of the launch vehicle, remain inside the launch vehicle airframe, be deployed and remain attached to the launch vehicle throughout the flight, or be deployed on the ground

9.1.2 Deployable payloads not remaining attached to the launch vehicle airframe, where separation is done prior to landing, shall:

9.1.2.1 Incorporate an independent recovery system, reducing the payload's descent velocity to less than 11 m/s (36 ft/s) at touchdown.

9.1.2.2 Contains a COTS GPS tracking system that complies with Section 4.0.

9.1.3 Deployable payloads may not implement gliders, drones, balloons, or any other UAS system.

9.1.4 Rockets with deployed payloads shall contain no more than a single (1) independently recoverable body.

Note: Payloads that are deployed after landing are considered as not deployable payloads for the purposes of this section.

9.2 PAYLOAD RECOVERY SYSTEMS

9.2.1 Payloads shall implement recovery systems that comply with all the requirements outlined in Sections 6.1 through 6.17. Mentions of "rocket" shall be understood to be the payload.

9.2.2 Payload batteries shall conform to requirements in Section 6.17.

9.3 PAYLOAD ENERGETIC DEVICES

9.3.1 Payload energetics shall be limited to the Payload Recovery Systems (Section 9.2). All other forms and uses of pyrotechnics and other energetics are not permitted.

9.4 PAYLOAD IDENTIFYING MARKINGS

9.4.1 All payloads shall be labeled with the school name, Team ID number, and the assigned GPS tracking frequency.[10]

9.4.2 This label shall be duplicated on each part of the payload which could separate either as designed or accidentally.

9.5 PAYLOAD LOCATION

9.5.1 The payload on two-stage rockets shall be in the sustainer.

[10] In some cases, frequencies will not be assigned until the time of the Final Flight Safety Review, and the requirement to list the assigned GPS tracking frequency on the payload is waived.

10.0 Launch and Ascent Trajectory Requirements

10.1 Launch Azimuth and Elevation

10.1.1 Launch officials will set the launch azimuth elevation. Single-stage rockets are typically launched at an elevation angle of 84° ±2° and multi-stage flights are typically launched at 87° ±2°.

10.1.2 Range Safety Officers reserve the right to require certain vehicles' launch elevation be lower or higher if flight safety issues are identified during pre-launch activities.

10.2 Ascent Stability

10.2.1 Launch vehicles shall maintain a specified minimum stability margin through the flight, regardless of Cg movement and/or shifting center of pressure Cp location, from launch through the first recovery system deployment event.

10.2.1.1 For subsonic flights, the minimum stability margin is 7.5% of the length of the rocket.

10.2.1.2 For supersonic flights, the minimum stability margin is 10% of the length of the rocket.

10.2.2 For two-stage rockets, the above minimum stability margins shall be determined separately for the stack and for the sustainer.

10.2.3 Teams shall provide simulation results showing the stability margin in calibers through the flight where the simulation is performed with no wind. The initial stability, in percent, shall be provided.[11]

10.2.4 Regardless of the above stability results, Teams shall not design rockets with a fin span less than 0.8 calibers, based on the largest diameter tube in the rocket (or in the stage for two-stage rockets).[12]

10.3 Over-Stability

10.3.1 Launch vehicles shall not exceed a specified maximum stability margin.

10.3.1.1 The stability margin at launch shall not exceed 18% of the length of the rocket.

10.3.1.2 The stability margin through the flight shall not exceed 25% of the length of the rocket.

10.3.2 Over-stability will be evaluated based on the simulation results described in 10.2.3.

[11] The stability margin should be determined from changes in the center of gravity through the flight combined with changes in the center of pressure as typically provided from rocket simulation programs such as OpenRocket. The specified methodology for determining rocket stability is intended to provide IREC reviewers with a consistent approach for evaluating the basic stability of the rocket (i.e., that the rocket is likely to go "up" when launched). It is not intended to replace a more detailed evaluation of aerodynamic characteristics of the rocket that a team may wish to conduct

[12] The intent of the fin span rule is to discourage the use of unreasonably small fins, even if stability measurements show that the rocket is stable. Small fin spans are particularly discouraged if the fin is "shadowed" by a larger-diameter tube above the fins.

11.0 ESRA Provided Launch Support Equipment

11.1 ESRA-provided Launch Rails

11.1.1 All teams competing in the solids (COTS or SRAD) and two-stage categories shall use IREC supplied launch control systems.

11.1.2 ESRA shall provide launch rails for single-stage "solids" rockets measuring at least 5.2 m (17 ft) long, 1.5" x 1.5" (aka 1515) aluminum guide rails of the 80/20® type, see https://8020.net/.

11.1.3 ESRA shall provide launch rails for two-stage rockets measuring at least 6.4 m (21 ft) long.

11.1.4 Teams shall contact the Hybrid Pad Manager for detailed information on the rails available for hybrid and liquid rockets.

11.1.5 Rockets shall be loaded horizontally from the top of the guide rail and the rail then erected to the required launch elevation.

11.1.6 Once erected, the launch vehicle shall be supported vertically by a submerged mechanical stop (aka "standoff") in the rail.

11.1.6.1 The position of the mechanical stop may be adjusted.

11.1.6.2 Teams can provide their own standoff. Team-provided standoffs will be subject to safety review, must be clearly marked with the Team ID number, and must be removable with commonly available tools.[13]

11.1.7 Teams whose designs anticipate requiring a longer launch rail to achieve stability during launch shall provide their own launch pad and rail.

11.2 ESRA-Provided Launch Control System

11.2.1 The IREC will utilize a Wilson F/X Wireless Launch Control System consisting of one LCU-64x launch control unit and PBU-8w encrypted wireless pad relay boxes.

11.2.2 Solids and two-stage rockets must use the ESRA-provided launch control equipment.

11.3 Team-Provided Launch Support Equipment

11.3.1 Teams are allowed to provide their own launch pads or towers.

11.3.2 Team-provided launch pads or towers shall be subject to safety review by ESRA personnel prior to use. Pads or towers deemed unsafe will not be permitted. Teams in this situation will be allowed to use ESRA-provided launch pads.

11.3.3 Hybrid and liquid launch controllers shall include instrumentation for monitoring oxidizer fill level. This instrumentation shall function in potentially low relative humidity of the launch site (<5%). Instrumentation relying on condensation of water in the air is unlikely to function.

11.3.4 Frequency Coordination

11.3.4.1 All team-provided launch control systems utilizing wireless communication shall coordinate their operations with the Hybrid Pad Manager.

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[13] Standoffs may be designed using the types of fittings commonly used for 1515 rails. The rails used in the Solids pad area also have holes through the sides of the rail that can be used to attach standoffs. Details on the location and size of these holes will be provided at a later time. Team-provided standoffs are allowed primarily for the purpose of helping to avoid the use of wooden blocks to support rockets.

11.3.4.2 All team-provided launch control systems utilizing wireless communication shall be able to displace their antennas at least 30 meters in any direction from the launch point.

11.3.4.3 Teams should elevate any antennas to provide line of sight clearance above bushes and terrain

11.3.5 Ground Power

11.3.5.1 Ground power umbilicals are permitted.

11.3.5.2 Ground power equipment shall not be connected to any ESRA equipment.

11.3.5.3 Generators are prohibited. Hybrid teams that require the usage of generators may be exempt from this requirement after coordinating and obtaining explicit approval from the Hybrid Pad Manager on a case-by-case basis

11.3.5.4 Excessive noise from the ground power support equipment (affecting launch operations) is prohibited.

11.3.5.5 Teams shall collect ground power equipment as soon as practical after use (*i.e.*, the next time the pad bank opens for access).

11.3.5.6 Team ground power equipment shall be labeled with team number and contact information.

11.3.5.7 Wet cell lead acid batteries are prohibited.

11.3.5.8 Teams anticipating the use of Li-Po batteries for ground power shall provide details in their progress reports.

11.4 Operational Range

11.4.1 All team-provided launch control systems shall be electronically operated and have a minimum operational range of no less than 610 m (2,000 ft) from the launch rail.

11.4.2 The maximum operational range is defined as the range at which abort and launch may be commanded reliably.

11.5 Fault Tolerance and Arming

11.5.1 All team-provided launch control systems shall be at least single fault tolerant by implementing a removable safety interlock (*i.e.,* a jumper or key to be kept in possession of the arming crew during arming) in series with the launch switch.

11.5.2 Appendix C of this document provides general guidance on assuring fault tolerance in high power rocketry launch control systems.

11.5.3 Hybrid/Liquid launch controllers shall include a failsafe mechanism providing automatic venting of oxidizer in the event of power or communication loss during fill.

11.6 Safety Critical Switches

11.6.1 All team-provided launch control systems shall implement ignition switches of the momentary, normally open (aka "deadman") type so that they will remove the signal when released.

11.6.2 Mercury or "pressure roller" switches are not permitted anywhere in team-provided launch control systems.

APPENDIX A: ACRONYMS, ABBREVIATIONS AND TERMS

ACRONYMS & ABBREVIATIONS	
ACS	Attitude Control System
AGL	Above Ground Level
APCP	Ammonium Perchlorate Composite Propellant
AWG	American Wire Gauge
CAS	Control Actuator System
CFR	Code of Federal Regulations
Cg	Center of Gravity
CONOPS	Concept of Operations
COPV	Composite Overwrapped Pressure Vessel
COTS	Commercial Off-the-Shelf
Cp	Center of Pressure
EIRP	Equivalent Isotropic Radiated Power
ESRA	Experimental Sounding Rocket Association
FoR	Flyer of Record
FRP	Fiber Reinforced Plastic
IREC	International Rocket Engineering Competition
LOX	Liquid Oxygen
MCOTS	Modified Commercial Off The Shelf
MSL	Mean Sea Level
NFPA	National Fire Protection Association
SBP	Student Band Plan
SRAD	Student Researched & Developed
TBD	To Be Determined
TBR	To Be Resolved
TRA	Tripoli Rocketry Association

TERMS	
Amateur Rocket	14 CFR, Part 1, 1.1 defines an amateur rocket as an unmanned rocket that is "propelled by a motor, or motors having a combined total impulse of 889,600 Newton-seconds (200,000 pound-seconds) or less, and cannot reach an altitude greater than 150 kilometers (93.2 statute miles) above the earth's surface".
Body Caliber	A unit of measure equivalent to the diameter of the launch vehicle airframe in question.
Excessive Damage	Excessive damage is defined as any damage to the point that, if the systems intended consumables were replenished, it could not be launched again safely. Intended Consumables refers to those items which are – within reason – expected to be serviced/replaced following a nominal mission (*e.g.,* propellants, pressurizing gases, energetic devices), and may be extended to include replacement of damaged fins or boat tails, specifically designed for easy, rapid replacement if such components are on hand and can reasonably be replaced within 30 minutes.
FAA Class 2 Amateur Rocket	14 CFR, Part 101, Subpart C, 101.22 defines a Class 2 Amateur Rocket (aka High Power Rocket) as "an amateur rocket other than a model rocket that is propelled by a motor or motors having a combined total impulse of 40,960 Newton-seconds (9,208 pound-seconds) or less."
Non-toxic Propellants	For the purposes of the IREC, the event organizers consider ammonium perchlorate composite propellant (APCP), potassium nitrate and sugar (aka "rocket candy"), nitrous oxide, liquid oxygen (LOX), kerosene, propane and similar, as non-toxic propellants. Toxic propellants are defined as requiring breathing apparatus, special storage and transport infrastructure, or extensive personal protective equipment, *etc.* Hydrogen peroxide is considered a toxic propellant.[14]
TRA Adult Flier	A TRA-insured flier who is 18 years old or older

[14] As indicated in the table, hydrogen peroxide is considered a toxic propellant. The current Rules & Requirements document indicates hydrogen peroxide as non-toxic. This DTEG designation for hydrogen peroxide will be used for the 2026 IREC.

APPENDIX B: SAFETY CRITICAL WIRING GUIDELINES

Introduction

- With the aim of supporting recovery reliability and overall safety, this Appendix sets out guidelines for all safety critical wiring. This is defined as wiring associated with recovery system deployments, any air-start rocket motors, location beacons (*e.g.,* GPS), and any systems that can affect the rocket trajectory or stability.
- The wiring techniques described here are suggested for inspection and ease of field repair. While non-critical wiring is outside the scope of this white paper the practices described herein are recommended.

Hazard Identification

- Any electronic system that utilizes voltages 50 volts (AC or DC) or higher shall be identified with the following warning external to the electronics enclosure; "CAUTION: High voltage inside. Internal access by team personnel only."
- Teams shall provide the rationale, in their progress reports, for the use of voltages exceeding 24 volts.

Wiring Guidelines

- All airborne wiring shall be stranded, insulated, 22 American Wire Gauge (AWG) or lower numerical AWG (the lower the AWG wire gauge the larger the conductor diameter) as appropriate (based on industry practice) to conduct expected currents. Copper conductors are required except as noted below.
  - Copper conductors plated with either silver or tin (entire wire, not just the ends) are desirable but not required.
  - When an off-the-shelf component includes flying leads, those leads may be used unmodified. For example, an E-match may contain solid wire, a battery connector may integrate 26 AWG wire, *etc.*
  - Stranded wire of a smaller diameter than 24 AWG may be used only when needed by an off-the-shelf component. For example, if the terminal block on an altimeter is sized to accept 24 AWG wires then that is the size of wire that should be used for that portion of the circuit.
- Wire should be stripped only with a wire stripping tool of the correct gauge. Any severed strands are cause for rejection.
- Each end of a wire should be terminated in one of the following approved methods:
  - Crimped into a crimp terminal (preferred). This includes crimp terminals on multi-conductor connectors such as 9-pin D-sub connectors (see table below).
  - Screwed into a binding screw terminal (acceptable).
  - Wires should be terminated into a terminal block, only if a piece of off-the-shelf equipment (*i.e.,* an altimeter) has built-in terminal blocks and so there is no other choice. Two-piece terminal blocks must be positively secured together – friction fit is insufficient.

  - Removal of wire strands to allow a wire to fit into a smaller hole or terminal is prohibited. Consider a smaller diameter conductor or different terminal type.
  - Wires will be terminated by soldering only if a piece of off-the-shelf equipment (*e.g.,* an arming switch) has built-in solder terminals and so there is no other choice. The reliability of solder joints cannot be fully established by visual inspection alone.
  - All crimp operations shall be performed with the correct tooling, using crimp terminals sized for the appropriate wire gauge.
  - Where multiple wires are crimped into a single terminal, calculate the effective gauge (for example, two 22 AWG are effectively 19 AWG) and use the appropriately sized terminal.
- Terminals with insulated plastic sleeves (usually color-coded to indicate barrel size) should be avoided because of the difficulty of inspection.
  - If a terminal is supplied with an insulated plastic sleeve it may be removed and clear heat shrink applied instead.
  - Insulated plastic sleeves on terminals are typically not required if they are mounted properly in barrier blocks. If insulation is needed, use clear heat-shrink tubing.
- When a bare wire is held down by a binding screw terminal the wire shall make a 180-degree hook, and strands shall be visible exiting the screw head. Only one wire shall be permitted per screw.
  - The wire bend shall be clockwise, so that it will tighten as the screw is torqued.
- When ring or spade terminals are held down by binding screw terminals, a maximum of two terminals are allowed per screw.
- A maximum of three wires shall be crimped into a single terminal barrel. Butt-splice terminals are considered to have separate barrels on each end.
- If two or more wires must be joined, one of the following approved methods should be used:
  - Crimp a ring terminal onto each wire, and then screw them into a barrier block. Add approved barrier block jumper pieces if many wires must be joined.
  - Screw the bare wire under a binding head screw in a barrier block. Add approved barrier block jumper pieces if many wires must be joined.
  - Crimp the wires into an uninsulated butt-splice terminal, and then insulate with clear heat-shrink tubing.
- Any wire-twisting splice method (including wire nuts) is explicitly forbidden.
- All wire terminations shall be inspected before flight.
  - Inspections will be visual to determine compliance with this document.
  - All terminations will be examined for exposed conductors that may have a potential for electrical shorts.

    - All terminations will be pull tested by exerting a moderate pull on the wiring to verify that it remains attached.
- All insulating tubing (usually heat-shrink) shall be transparent. This allows inspection of the underlying hardware.
    - No tape, glue or RTV shall be used to insulate or bundle any element of the wire harness.

Connector/Harness Guidelines

- Connectors shall be identified by labels and/or color coding to allow determination of the associated system and mating parts.
- Connectors (except coaxial connectors) shall use crimp contacts unless no alternative can be procured.
    - Coaxial connectors may be soldered.
- Mated connectors (except threaded coaxial connectors) shall utilize a positive locking mechanism to keep the two halves mated under vibration and tension.
    - If the mated connectors do not have a positive locking mechanism built-in, they shall utilize an external means of preventing connection separation, *e.g.,* tie wraps, lacing, mechanical clamps.
    - Threaded coaxial connectors will be checked for finger tightness.
- Plastic connector latches are allowed but subject to inspection for damage and engagement.
- Power connectors shall be polarized to prevent reversed power application.
- Good practice recommends that individual wires be bundled together to make a harness (factory multi-conductor wiring in a common outer jacket is acceptable).
    - Safety critical harnesses shall be kept separate from payload harnessing (as applicable).
    - Bundling may be accomplished by:
        - A light twist (for mechanical reasons only, no EMC mitigation is intended).
        - Short (1 cm) lengths of clear heat-shrink tubing, zip-ties, or lacing every 5 cm.
        - Wire mesh sleeving, provided it allows for inspection of the wiring inside.
- The harness may be supported by plastic P-clamps, tie bases, or by tie wraps to structure. Harness routing shall not be in contact with sharp edges, *e.g.,* screw threads, that may damage wiring.
- All items that are connected by the harness (barrier blocks, sensors, batteries, actuators, switches, *etc.*) should be rigidly fixed to the rocket structure so that they cannot move.
    - Rigid fixing implies attachment with clamps using threaded fasteners or a solid glue bond. Cable ties and/or tape are not acceptable examples of rigid fixing.
    - All wiring should have strain relief (small amount of slack) to prevent terminations and connections from demating.
- Batteries should be mounted and connected appropriately:

- "Velcro" (or similar hook and loop material) and/or tie wraps shall not be used as the sole method of mounting batteries.
- Batteries that are subject to puncture damage should be mounted in a fashion and location to mitigate the likelihood of such damage during a hard landing or other high impact event.
- 9V alkaline batteries shall be secured to prevent motion and connected using proper snap terminals.
  - Batteries shall be mounted in a direction and manner that prevents acceleration forces from disconnecting the battery connector.
- Gel-cell lead acid battery connections shall be color coded (typically red for positive, black for negative) to avoid reversed connection polarity. Insulated "faston" terminals may be used in this application subject to visual inspection and pull testing.
  - Gel cell batteries, because of their weight, will be inspected for rigid and captive mounting. Tie wraps and/or Velcro hook and loop alone is not acceptable.
- Cylindrical batteries (AAA, AA, C, D, 18650, *etc.*) may be mounted into commercial holders. Battery holders, if used, shall be rigidly secured to the structure, and the batteries shall then be strapped (*e.g.,* tie wrap, lacing) into the holders.
  - Battery holders shall be mounted in a direction and manner that prevents acceleration forces from compressing contact springs.

Circuit Board Guidelines

- All components greater than 7g shall be staked to prevent motion under acceleration.
  - The use of conformal coatings is suggested to provide a degree of staking for all components and to avoid electrical shorts.
- All socketed components (*e.g.,* integrated circuits) and press-fit contacts shall be positively restrained so that they cannot come loose under vibration.
- Wire wrap, through-hole solder, and surface-mount solder are acceptable fabrication methods.
- Solderless breadboards (also known as plug-in breadboards) shall not be used.
- Any commercial board for the high-power rocketry market should be considered to be of sufficient quality, provided it is in an undamaged factory state.
- The connector between the motherboard and daughterboards shall not be the sole means of support. Mechanical supports for the daughterboard are required.
- Cuts and jumpers are permitted if the jumpers are staked to prevent damage.

Recommended Sources

- Suggested components can be purchased from Digikey, Mouser, Omron, and Amazon that will help to satisfy the wiring guidelines.

- These are recommendations only, and you are free to choose other parts and buy from other suppliers.
- Look up the catalog pages associated with each Digikey or Mouser number to find similar parts of different sizes.

PART	NUMBER	NOTES
Wire	Digikey A5855W-100-ND	This is good 22-gauge, tinned, Teflon insulated wire. Cold-flow is a long-term consideration, but shouldn't be a problem for a short lifetime rocket.
Wire	Digikey C2016L-100-ND	22-gauge tinned PVC-insulated wire. Note that the "L" designates the insulation color (other colors are B,R,A,Y,N,W)
Wire	Digikey W120-100-ND Digikey W121-100-ND	2-conductor, 22-gauge 3-conductor, 22-gauge
Wire	Amazon "Tinned marine grade wire"	18-gauge, available in 35-ft or 100-ft rolls
Ring terminals, uninsulated	Digikey A27021-ND (#6 hole)	The Solistrand series is a high quality terminal. Various crimp tools are available. You get what you pay for – the expensive ones are very nice, but the basic ones will do in a pinch.
Butt-splice terminal	Digikey A09012-ND	Another Solistrand series terminal
"Faston" terminal	Digikey 298-10011-ND (check size)	These terminals are useful for connecting switches, gel cell batteries, and many automotive devices
9V battery holder, with solder terminals	Digikey 708-1409-ND	Screw this holder to your chassis, and then cable tie the battery in.
9V battery connector, snap on	Digikey BS12I-ND	Cable tie connector to battery and battery to chassis.
4 AA battery holder	Digikey 708-1399-ND	This is a nice enclosed battery box for 4 AA cells
P-clamp	Digikey 7624K-ND (check size)	This particular unit is for a 0.25" dia. Harness. Select the correct size.

PART	NUMBER	NOTES
Heat-shrink tubing	Digikey A014C-4-ND (check size) Mouser 650-RNF100 (check size)	Material is clear polyolefin with low shrink temperature. Shrink with hot-air gun or oven.
Barrier block (double row)	Digikey CBB206-ND Mouser 538-2140 or 4140 (0.375" pitch), 538-2141 or 4141 (0.438" pitch)	Available in a range of lengths. Can accept ring or spade terminals (preferred), or bare wire (acceptable).
Barrier block jumper	Digikey CBB314-ND	Connect adjacent strips when many wires need to be connected together
D-sub connectors (9 contact)	Digikey A31886-ND (male shell) Digikey A34104-ND (female shell) Digikey A1679-ND (male pins) Digikey A1680-ND (female pins)	The connectors and contacts are cheap, but the crimp tools are expensive.
D-sub fixing hardware	Digikey MDVS22-ND (screw) Digikey MDVS44-ND (socket)	These kits convert the D-sub friction fit into a proper positive lock.
MIL-C-38999 connectors	Digikey 956-1017-ND (13 pin panel mount receptacle with pins) Digikey 956-1020-ND (13 pin plug with sockets)	These connectors approach the style and quality used on orbital launch vehicles. Extremely robust, but very expensive!
Switch for pull-pin	Omron SS-5G	This switch is rated to 30G. Available direct or as part of some commercial pull-pin switches

APPENDIX C: FIRE CONTROL SYSTEM DESIGN GUIDELINES

Introduction

- The following Appendix is written, for hybrid and liquids teams, to illustrate safe fire control system design best practices and philosophy to student teams participating in the IREC. When it comes to firing (launch) systems for large amateur rockets, safety is paramount. This is a concept that everyone agrees with, but it is apparent that few truly appreciate what constitutes a "safe" firing system. Whether they've ever seen it codified or not, most rocketeers understand the basics:
- The control console should be designed such that two deliberate actions are required to fire the system.
- The system should include a power interrupt such that firing current cannot be sent to the firing leads while personnel are at the pad and this interrupt should be under the control of personnel at the pad.
- These are good design concepts and if everything is working as it should they result in a perfectly safe firing system. But "everything is working as it should" is a dangerous assumption to make. Control consoles bounce around in the backs of trucks during transport. Cables get stepped on, tripped over, and run over. Switches get sand and grit in them. In other words, components fail. As such, there is one more concept that should be incorporated into the design of a firing system.
- The failure of any single component should not compromise the safety of the firing system.

Proper Fire Control System Design Philosophy

- Let us examine a firing system that may at first glance appear to be simple, well designed, and safe (Figure 1). If everything is functioning as designed, this is a perfectly safe firing system, but let's examine the system for compliance with proper safe design practices.
- The control console should be designed such that two deliberate actions are required to launch the rocket. Check! There are actually three deliberate actions required at the control console: (1) insert the key, (2) turn the key to arm the system, (3) press the fire button.
- The system should include a power interrupt such that ignition current cannot be sent to the firing leads while personnel are at the pad and this interrupt should be under control of personnel at the pad. Check and check! The Firing relay effectively isolates the electric match from the firing power supply (battery) and as the operator at the pad should have the key in his pocket, there is no way that a person at the control console can accidentally fire the rocket.
- But all of this assumes that everything in the firing system is working as it should. Are there any single component failures that can cause a compromise in the safety of this system? Yes. In a system that only has five components beyond the firing lines and e-match, three of those components can fail with potentially lethal results.

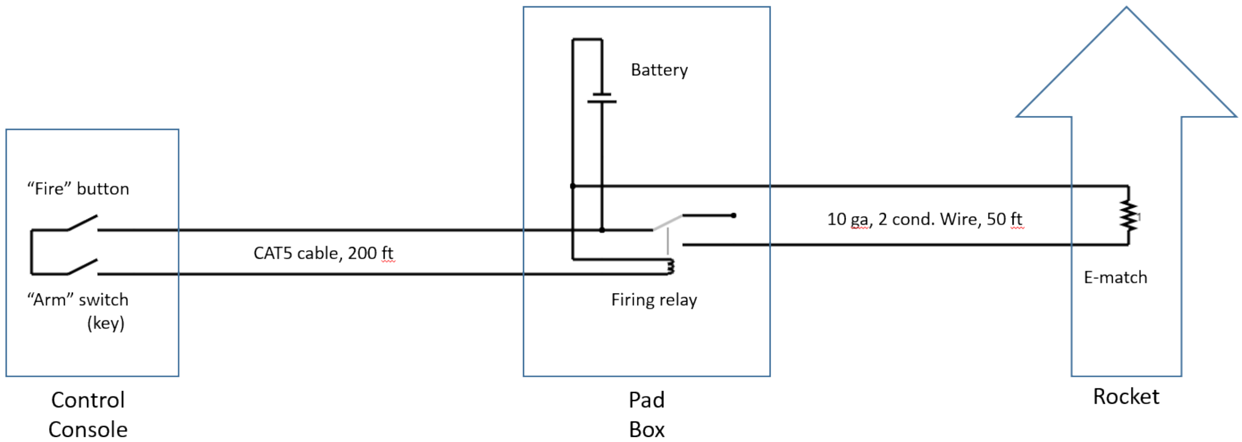


Figure 1: A simple high current fire control system.

Firing Relay

- If the firing relay was stuck in the ON position: The rocket would fire the moment it was hooked to the firing lines.
- This is a serious safety failure with potentially lethal consequences as the rocket would be igniting with pad personnel in immediate proximity.

Arming Switch

- If the arm key switch failed in the ON position simply pushing the fire button would result in a fired rocket whether intentional or not. This is particularly concerning as the launch key – intended as a safety measure controlled by pad personnel – becomes utterly meaningless.
- Assuming all procedures were followed, the launch would go off without a hitch. Regardless, this is a safety failure as only one action (pressing the fire button) would be required at the control console to launch the rocket. Such a button press could easily happen by accident.
- If personnel at the pad were near the rocket at the time we are again dealing with a potentially lethal outcome

CAT5 Cable

- If the CAT5 cable was damaged and had a short in it the firing relay would be closed and the rocket would fire the moment it was hooked to the firing lines. This too is a potentially lethal safety failure.

Notice that all three of these failures could result in the rocket being fired while there are still personnel in immediate proximity to the rocket. A properly designed firing system does not allow single component failures to have such drastic consequences. Fortunately, the system can be fixed with relative ease.

Consider the revised system (Figure 2). It has four additional features built into it: (1) A separate battery to power the relay (as opposed to relying on the primary battery at the pad), (2) a flip cover over the fire button, (3) a lamp/buzzer in parallel with the firing leads (to provide a visual/auditory warning in the event that voltage is present at the firing lines), and (4) a switch to short the firing leads during hookup (pad personnel should turn the shunt switch ON anytime they approach the rocket).

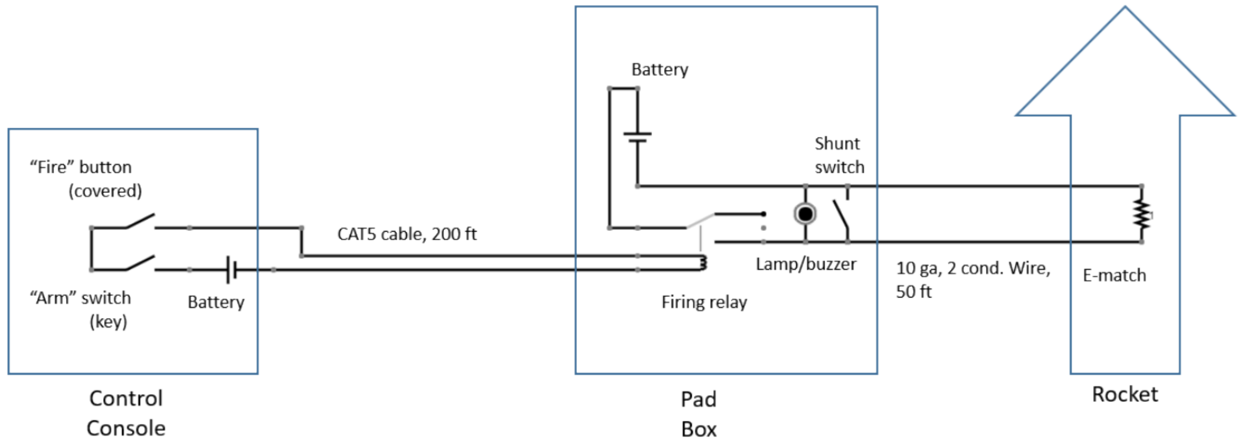


Figure 2: An improved high current fire control system.

In theory, these simple modifications to the previous firing circuit have addressed all identified single point failures in the system. The system has 8 components excluding the firing lines and e-match (part of the rocket itself). Can the failure of any of these components cause an inadvertent firing? That is the question. Let us examine the consequences of the failure of each of these components.

Fire Button

- If the fire button fails in the ON position, there are still two deliberate actions at the control console required to fire the rocket. (1) The key must be inserted into the arming switch, and (2) the key must be rotated.
- The firing will be a bit of a surprise, but it will not result in a safety failure as all personnel should have been cleared by the time possession of the key is transferred to the Firing Officer.

Arm Switch

- If the arm switch were to fail in the ON position, there are still two deliberate actions at the control console required to fire the rocket. (1) The cover over the fire button would have to be removed, and (2) the fire button would have to be pushed.
- This is not an ideal situation as the system would appear to function flawlessly even though it is malfunctioning and the key in the possession of personnel at the launch pad adds nothing to the safety of the overall system.

- It is for this reason that the shunting switch should be used. Use of the shunting switch means that any firing current would be dumped through the shunting switch rather than the e-match until the pad personnel are clear of the rocket.
- Thus, personnel at the pad retain a measure of control even in the presence of a malfunctioning arming switch and grossly negligent use of the control console.

Batteries

- If either battery (control console or pad box) fails, firing current cannot get to the e-match either because the firing relay does not close or because no firing current is available.
- No fire means no safety violation.

CAT5 Cable

- If the CAT5 cable were to be damaged and shorted, the system would simply not work as the current intended to pull in the firing relay would simply travel through the short. No fire means no safety violation.

Firing Relay

- If the firing relay fails in the ON position the light/buzzer should alert the pad operator of the failure before he even approaches the pad to hook up the e-match.

Shunt Switch, Lamp/Buzzer

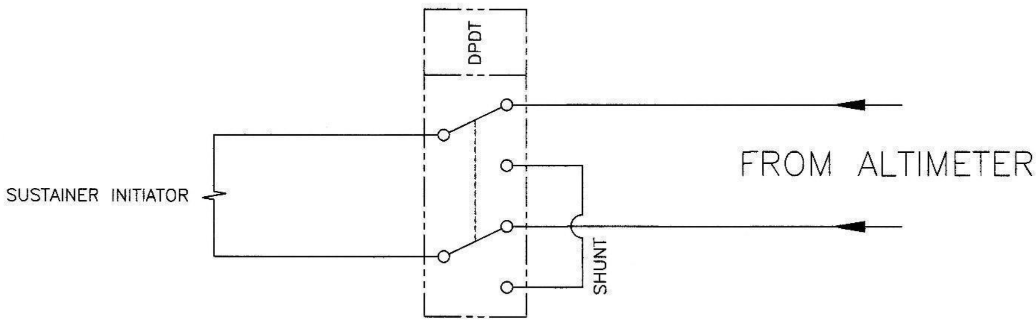
- These are all supplementary safety devices.
- They are intended as added layers of safety to protect and/or warn of failures of other system components.
- Their correct (or incorrect) function cannot cause an inadvertent firing.

Concluding Remarks

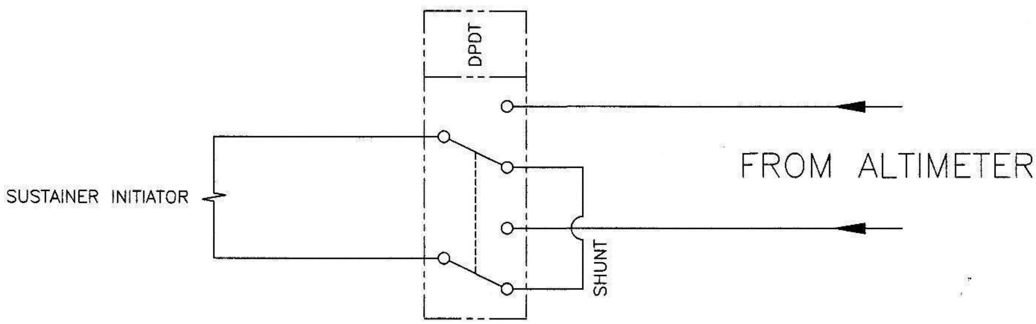
Is this a perfect firing system? No. There is always room for improvement. Lighted switches or similar features could be added to provide feedback on the health of all components. Support for firings at multiple launch pads could be included. Support for the fueling of hybrids and/or liquids could be required. A wireless data link could provide convenient and easy to set up communications at greater ranges. The list of desired features is going to be heavily situation dependent and is more likely to be limited by money than good ideas.

The circuit should be designed such that no single equipment failure can result in the inadvertent firing of the e-match and thus, the rocket motor. Whether or not a particular circuit is applicable to any given scenario is beside the larger point that in the event of any single failure a firing system should always fail-safe and never fail in a dangerous manner. No matter how complicated the system may be, it should be analyzed in depth and the failure of any single component should never result in the firing of a rocket during an unsafe range condition. Note that this is the bare minimum requirement; ideally, a firing system can handle multiple failures in a safe manner.

APPENDIX D: AIR START IGNITION WIRING DIAGRAM



SUSTAINER INITIATOR ARMED



SUSTAINER INITIATOR SAFE

Appendix E: ESRA & Tripoli Rocket Association Partnership

Summary

The Experimental Sounding Rocket Association (ESRA) and the Tripoli Rocketry Association (TRA) are now formally working together to continue to improve overall flight safety and efficient flight operations at the annual IREC. Going forward, all operations and all flights at the IREC, whether competitive or demonstration, will be conducted under the TRA Unified Safety Code. Details of how these impacts competing teams are described below.

Background

TRA and ESRA share common goals to create safe and exciting launch opportunities for the next generation of Aerospace engineers as they progress from hobby to industry environments. Our organizations are highly aligned: ESRA provides the educational framework and administration of the world's largest international collegiate rocketry event; and TRA provides the safety/flight operations framework, incredible membership expertise and an amazing insurance program. Stated simply, we have assembled outstanding partners to build and safely operate the greatest rocketry competition in history.

The IREC has relied on the TRA Unified Safety Code and membership expertise/guidance to Create a safe launching environment for its event since 2017. IREC's Range Safety and Launch Operations teams are filled with a significant number of Senior Level 3 high power rocketry experts (>30). A growing number of competing teams utilize TRA prefectures for mentorship, certification flights and test flights of their competition projects.

Benefits:

- **Student Team Liability Insurance** – In previous years, students were either insured by their university or through a 3rd party insurance provider. By becoming a TRA student member and following the TRA Unified Safety Code, the student is insured by TRA's insurance program. TRA insurance covers COTS and SRAD solid and hybrid flights, as well as liquid engine flights that conform to the requirements in the Safety Code.
- Student teams will have improved access to TRA mentors, certifications, and launch sites – TRA has a growing list of US and international prefectures with active launch sites. Student teams are highly recommended to engage with their local TRA prefectures for mentoring, certification flights and additional launch experience.

Tripoli Flight Requirements for All Competition Categories:

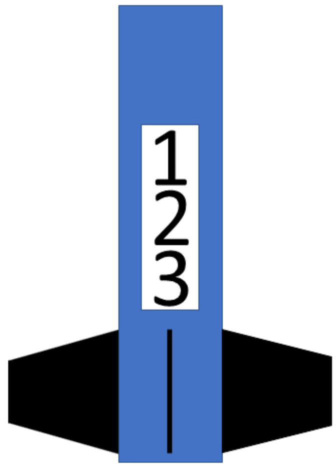
- A TRA-certified Level 2 or Level 3 Flyer of Record (depending upon motor impulse) shall be required for all competition categories and must be present for launch preparation, pad loading and recovery activities.[15] There are three options to satisfy the requirement for a Level 2/3 certified Flyer of Record.
    1. Recommended: Each team will secure a TRA Level 2/3 certified Adult Flier who works closely with the team and is present as the Flyer of Record for the launch preparation, pad loading and recovery activities. Student teams ARE ENOURAGED to subsidize the travel expenses of their Flyer of Record, both to and from the event.
    2. Also recommended: A student, who is TRA-certified Adult Flier, and who has significant rocketry experience, can serve as the FoR. The student FoR must be onsite with the team for all aspects of the review and launch.
    3. International Teams should begin their FoR search immediately and utilize social media platforms, forums and other community outlets to secure a Flyer of Record. ESRA WILL NOT BE PROVIDING L2 or L3 FoR's.
- TRA membership is required to be on the range or to work on safety-related systems on the rocket.
- Motors or energetic materials may only be possessed or handled by TRA members with appropriate high power rocketry certification.
- Student teams are required to meet all TRA launch safety requirements to ensure TRA insurance covers their flight and their participation at the IREC.

TRA Student Membership

- TRA Student membership is $20 per year per student. This is a requirement for ANY student who will be at the launch site.
- A maximum of 5 additional team members who are TRA student members may be on the pad loading team. These student team members do not have to be HPR certified but must be TRA student members. All students are encouraged to secure HPR certification.

[15] As described in Section 3.1, an FoR with a Level 3 certification is required for flights that include an M, N or O motor or a combination of motors exceeding a total impulse of 5,120Ns. An FoR with a Level 2 certification is acceptable for flights that include a K or L motor or a combination of motors having a total impulse of less than 5,120Ns.

APPENDIX F: EXAMPLE BOOSTER AIRFRAME MARKINGS



IREC Team ID Number 123 on a 4 fin, 6-inch diameter rocket

APPENDIX G: STANDARD WEATHER CONDITIONS

The following weather conditions are standard at the launch site:

WEATHER PARAMETERS		
Temperature	Average High: 33-35 °C (92-95 °F) Average Low: 19-22 °C (67-72 °F)	Can get as high as 42° C (108° F)
Windspeed	Average: 18.3-19.0 km/h (11.4 – 11.8 mph)	Out of the South (77% of the time)
Precipitation	Rain Probability: 21-25% on any given day Average: 1.5-2.0 ” total for the month	Occasional heavy rain that can affect road access to the launch area.
UV Index	Average High: 7 Average Max: 11	
Dust	Everywhere	
Background Noise at Pad	76 dBA (Typical)	ESRA generator with no wind

APPENDIX H: DOCUMENT REVISIONS

REVISION	DESCRIPTION	DATE
2026 V1.1	**Minor Revision for 2026 IREC Competition** There are numerous edits throughout the document intended to improve the clarity of the document. ESRA recommends that teams carefully review any sections that may affect their participation in the 2026 IREC. Listed section numbers refer to the section numbers in this document and not the prior version. **Modify 3.1**. Modified to clarify that a TRA Adult Flier who is certified to Level 2 can serve as an FoR if the motors to be flown in the rocket are K or L motors, where the total impulse of all motors in the rocket is less than 5,120Ns. **Added 4.1**. References the Student Band Plan as the source of radio transmission information for the IREC, and includes a link to the plan. **Added 4.3.4.** Specifies how frequencies will be assigned within the COTS GPS frequency ranges. **Modified 4.3.5.** Specifies allowable radio transmissions in the pit area. **Added 4.3.11.** Includes a link to the rules and procedures for the Live Video Challenge. **Added 5.14.3.1.** Allows devices that will provide power to flight computers (e.g., WiFi switches) to be powered up before the rocket is taken to a vertical position. **Added 6.1.7.** Explicitly prohibits the use of systems that “guide” a rocket or payload on descent. **Added 8.6.11.** Allows exceptions to the rail button mounting requirements for minimum diameter rockets, subject to ESRA approval. **Modify Appendix E.** Clarifies that all IREC activities and flights are performed under the TRA Unified Safety Code.	02/23/2026

REVISION	DESCRIPTION	DATE
2026 V1.0	**Initial release for 2026 IREC Competition** There are numerous edits throughout the document intended to improve the clarity of the document. ESRA recommends that teams carefully review any sections that may affect their participation in the 2026 IREC. Listed section numbers refer to the section numbers in this document and not the prior version. **Amend 1.2.3**. References the use of Discord and not HeroX for IREC communications. **Modified 4.1.1.** Modified the table of COTS GPS solutions to recommend solutions that can be tracked by MCC. **Added 4.1.2**. Specifies the requirements for a Qualified COTS GPS transmitter. **Clarify 4.1.3**. Clarifies that tracking solutions will be both *switched* and powered independently of any flight computer. **Amend 4.2.** This section is amended to specify the development of COTS GPS frequency ranges and the procedures that will be used to assign COTS GPS frequencies to teams that do and do not have access to HAM licenses or reciprocal HAM license agreements. **Add 4.2.2.4.** Specifies requirements for Qualified SRAD GPS and other SRAD transmitters. **Amend 4.2.3**. Defines procedures for using multiple COTS GPS solutions in a rocket or a section of a rocket. **Amend 4.2.4.** Specifies procedures for testing GPS transmitters in the pits. **Amend 4.2.8.** Amended for a minor change in the assignment of custom names for GPS trackers. **Added 4.2.10.** Specifies the use of 1.2, 2.4 and 5.8GHz for hybrid/liquid ground support systems. **Amend 4.3.2.** Amended to recommend the use of APRS protocol via 1200 baud AFSK for SRAD GPS transmitters. **Clarify 5.1.2.2.** Clarifies that commercial propellant grains cannot be used in SRAD motors. **Clarify 5.1.2.3.** Clarifies that SRAD motor testing and propellant characterization must be completed prior to the submission of the Technical Report.	10/15/2025

REVISION	DESCRIPTION	DATE
	Added 5.2.3.4. Specifies that the use of oxidizers other than nitrous oxide must be reviewed by ESRA. **Amend 5.3.1.** Specifies a revised minimum rail exit velocity of 25 m/s and defines the procedure for determining "effective" rail length. **Added 5.3.3.** Defines procedures for determining the rail exit velocity for tower launchers. **Amend 5.4.3.** Requires double-headed igniters for ground-start motors and lists other igniter requirements. **Amend 5.5**. Clarify that hydrogen peroxide is considered a toxic propellant. **Amend 5.14.2.** Requires the use of a shunt on air-start motors. **Added 5.14.2.4**. Includes information on allowable interactions between SRAD electronics and air-start motor igniter circuits. **Clarify 5.14.3.** Clarifies that flight computers cannot be "powered up" until the rocket is vertical. **Amend 6.1.3.** Requires dual-deploy recovery on two-stage boosters. **Remove** footnote specifically recommending the use of a Blue Raven or a separate Blue Raven as the official altitude logging altimeter. **Added 6.15.6.** Documents anticipated ladders at the solids and two-stage pads. **Amend 6.17.1.1.** Prohibits the use of Li-Po batteries for rockets and payloads. **Amend 6.19** Prohibits the use of >200mw transmitters in the 2m, 70cm and 33cm bands. **Added 8.2.3** Requires teams to provide the value of the shear modulus used in their fin flutter calculation. **Amend 8.3.1.** References sections of the Tripoli Unified Safety Code dealing with the use of metal in rockets. **Various sections.** Added reporting requirements for rocket design features such as rail buttons, shear pins, and venting. **Amend 10.2.** Includes specific requirements for minimum stability subsonic and supersonic flights. **Added 10.2.4** Prohibits fin semi-spans less than 0.8 calibers. **Amend 10.3.** Includes specific requirements for maximum stability on the pad and during the flight.	

REVISION	DESCRIPTION	DATE
	Added 11.1.5 Allows teams to provide custom rail standoffs. **Modified 11.3.5.** Requires teams to notify ESRA is using LiPo batteries in their ground power systems.	